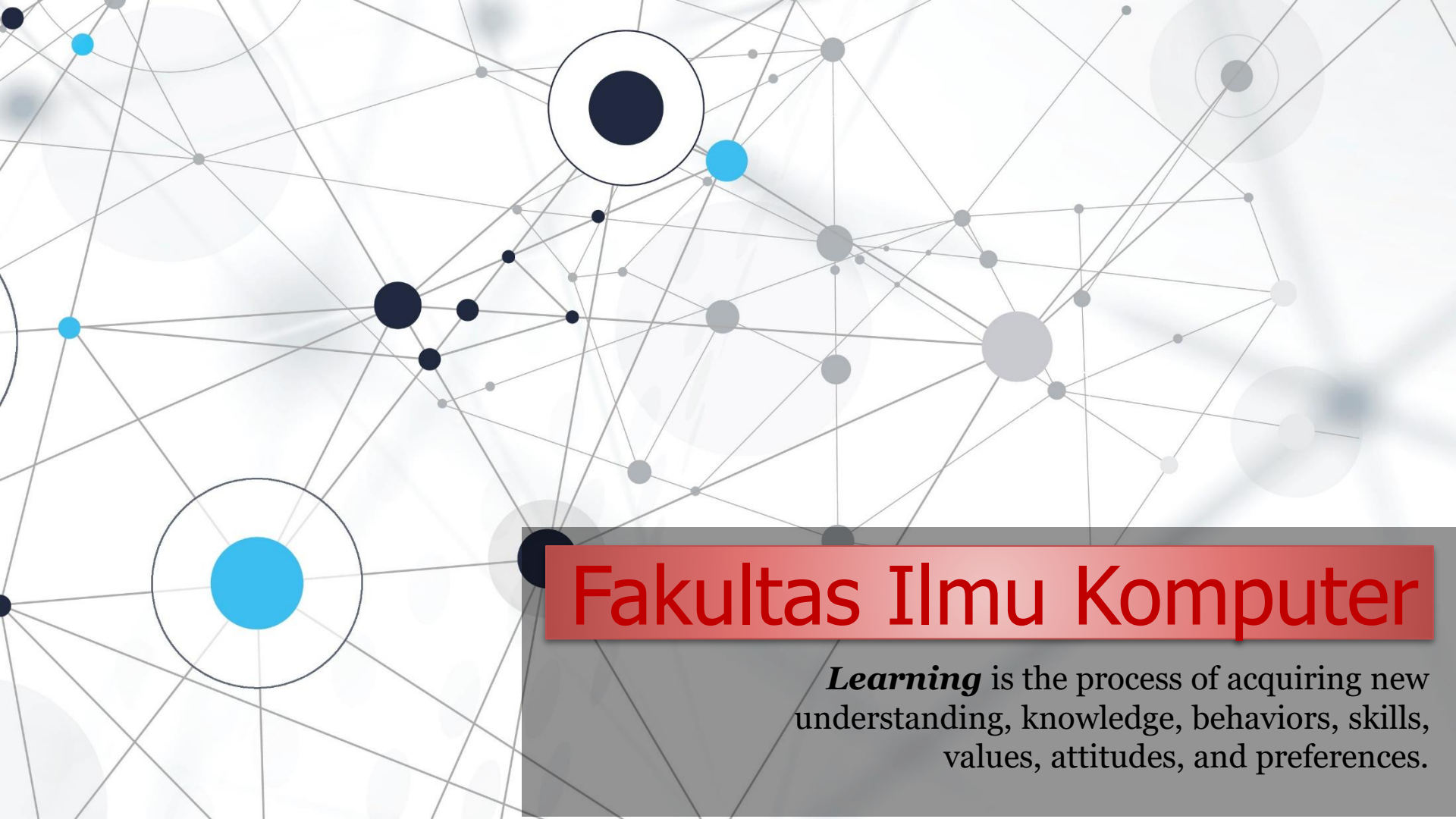


An abstract network diagram with various sized nodes (black, blue, and grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

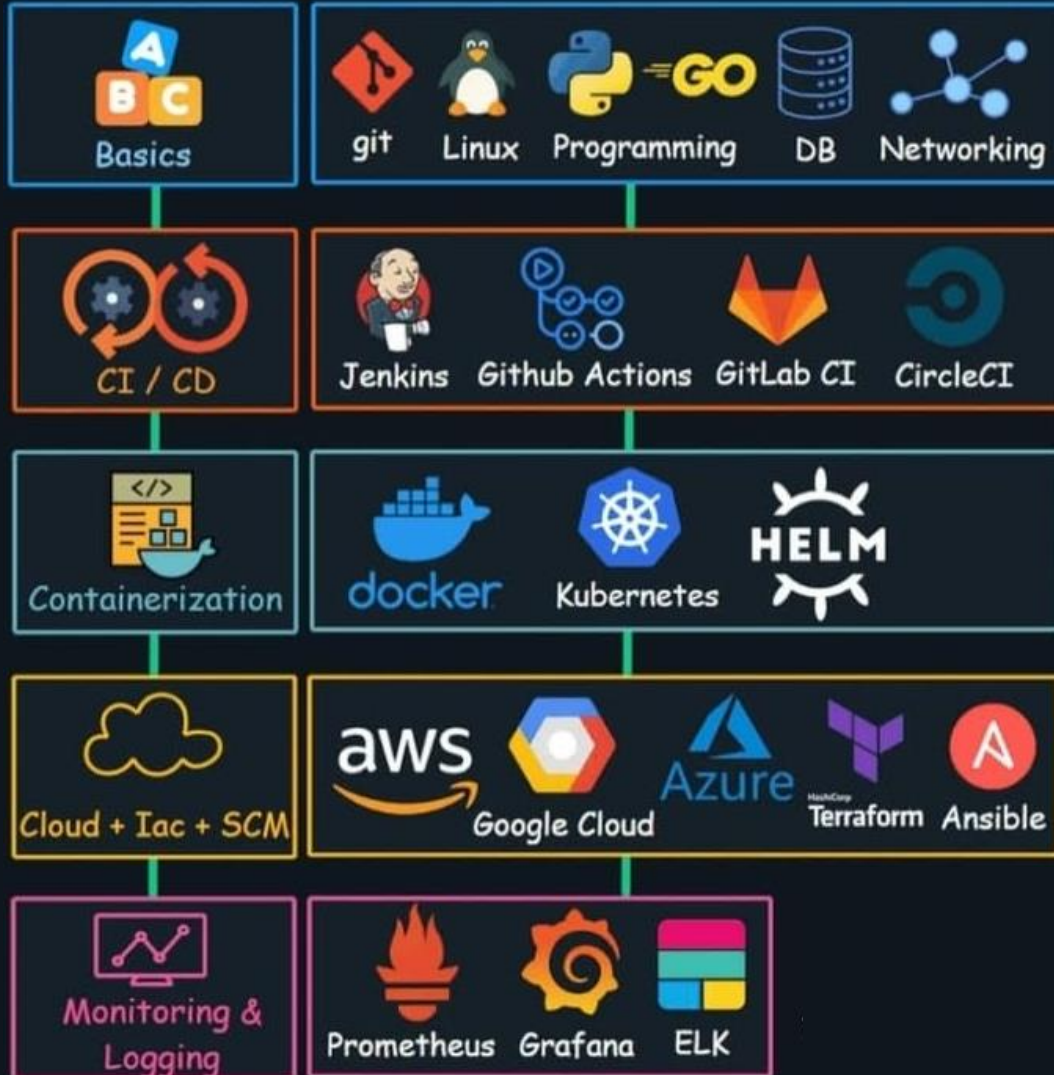
Monitoring & Logging *in* Cloud Server



Welcome to *the* DevOps Engineering *course*

DevOps Learning Path

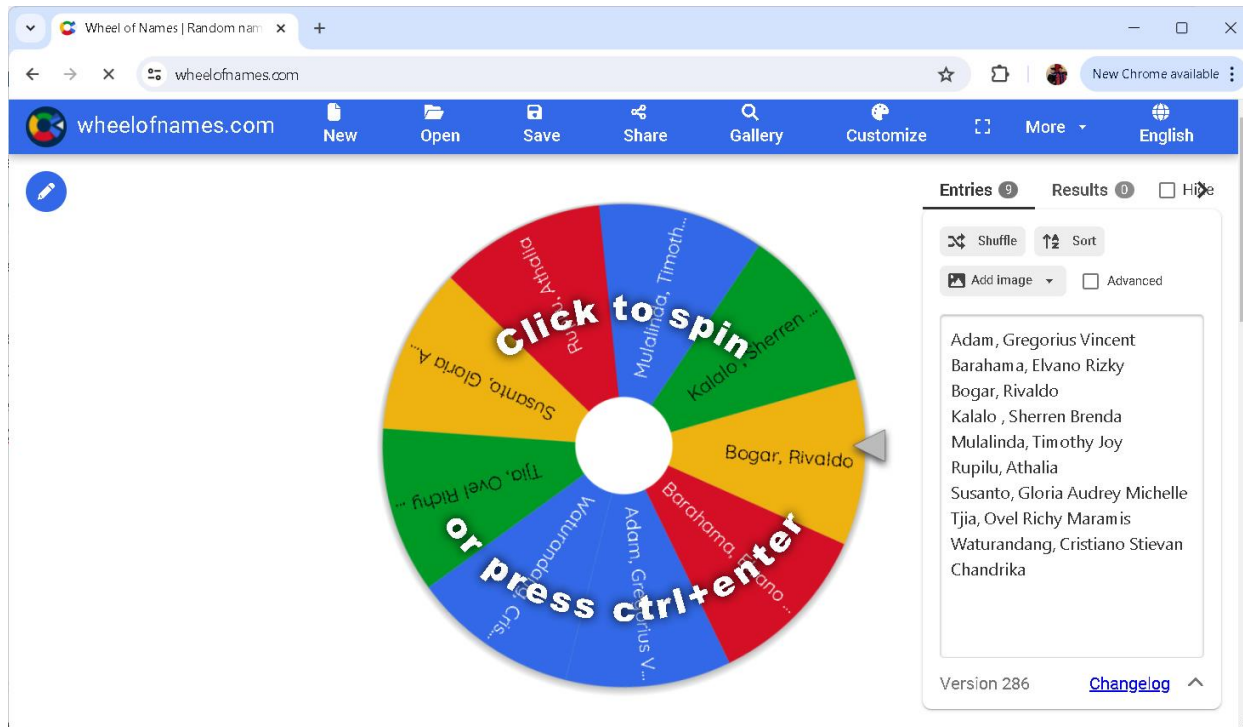
For **Beginners**



Typical SCM process:

(React, Vue, Svelte, atau Angular)

- 1) Developer **clones** a Git repository.
- 2) Makes changes to code.
- 3) **Commits** the changes locally with a message.
- 4) **Pushes** changes to a remote repository (e.g., GitHub).
- 5) A **Pull Request (PR)** or **Merge Request (MR)** is reviewed by teammates.
- 6) Once approved, changes are merged into the main branch.
- 7) The CI/CD pipeline (e.g., Jenkins, GitHub Actions) automatically builds and deploys the new version.



Random Picker:
<https://wheelofnames.com/>



Apakah manfaat dalam monitoring server (fisik dan cloud)?



Manfaat dari adanya kamera pengawas???

Banyak negara, termasuk Taiwan, menggunakan CCTV untuk mengawasi area publik, pusat perbelanjaan, stasiun, dan sebagainya.



Security

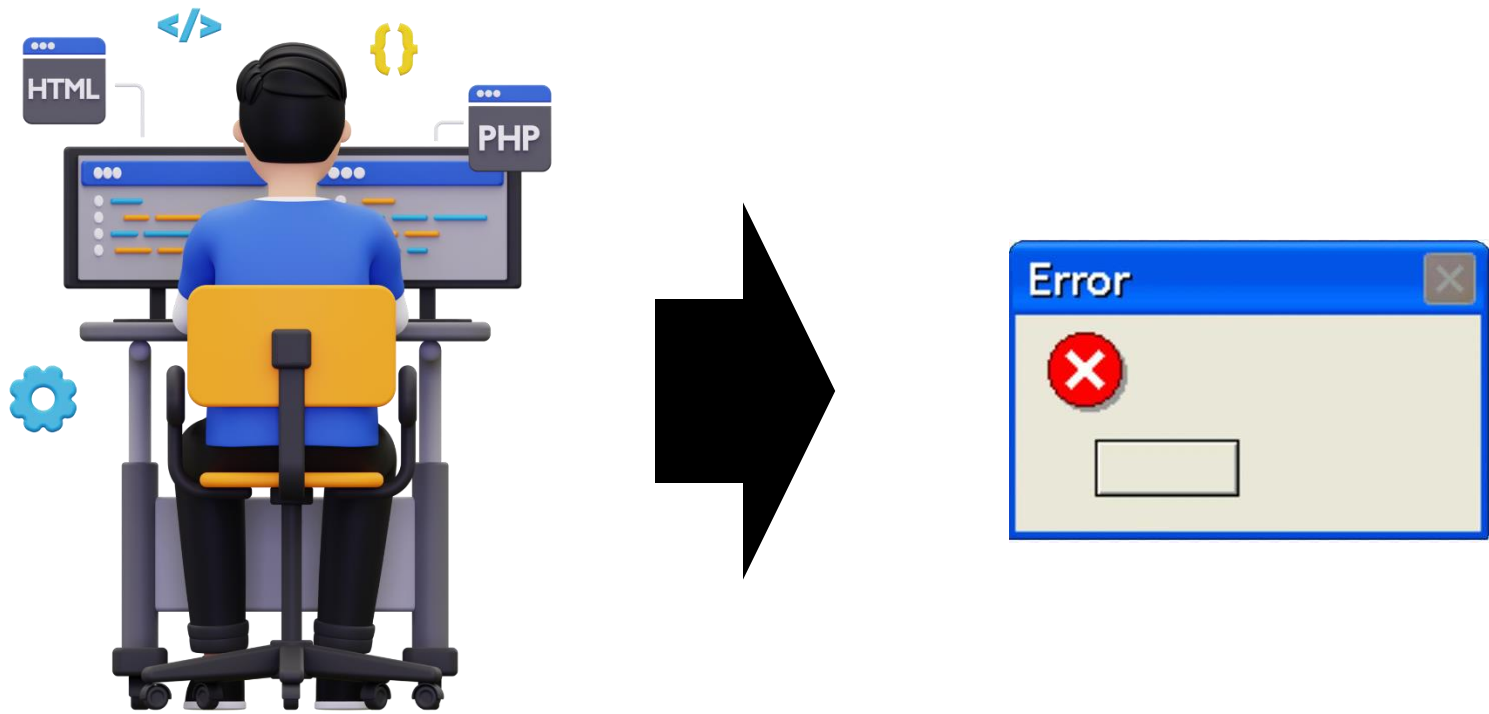
Monitoring

A photograph of a server rack with a white sign taped to the front. The sign has the text 'IN CASE OF CYBERATTACK' and 'BREAK GLASS AND PULL CABLES' in bold, black, sans-serif capital letters. The server rack is open, revealing a dense bundle of blue and white network cables plugged into the back of the server units. To the left of the rack, on a white shelf, there is a green plastic watering can and a yellow container.

**IN CASE OF
CYBERATTACK**

**BREAK GLASS
AND PULL CABLES**

Is this real?
When cyberattacks
occur, the first
thing to do is to put
the company
offline.



Sebagai seorang *programmer*, penting untuk memiliki *sistem alarm* yang efektif untuk melacak dan menangani kesalahan (*error*) dalam program.

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1 - Server Monitoring *vs* Cloud Monitoring

A decorative border of green leaves and branches at the bottom of the slide.

Why Monitor & Log in DevOps?



Monitoring and logging are essential for DevOps in cloud computing as they enable us to detect and troubleshoot issues before they impact customers or users, optimize the performance and availability of cloud resources and applications, analyze and improve DevOps processes and workflows, comply with regulatory and security standards and policies, as well as gain insights from data and metrics.

Monitors and logs are two critical components in DevOps practices as they are used to monitor, measure, and analyze application and infrastructure performance in development, test, and production cycles.

Monitor & Log DevOps

How can you monitor and log effectively when implementing DevOps in Cloud Computing?

- ❖ DevOps is a set of practices that aims to **improve the collaboration** and efficiency of software development and operations teams.
- ❖ Cloud computing is a model that provides on-demand access to scalable and flexible IT resources over the internet.
- ❖ When you **combine DevOps and cloud computing**, we can achieve *faster delivery, higher quality, and lower costs* of our software products.
- ❖ To ensure the *reliability* and *security* of our **cloud-based DevOps pipelines**, **we need to monitor and log our processes** and systems effectively.

Monitor & Log DevOps

Monitors can provide early notification of potential problems before they affect end users.

Monitor dapat memberikan pemberitahuan dini tentang potensi masalah sebelum mereka memengaruhi pengguna akhir.

Logs are a valuable source of information for diagnosing problems and tracking application activity.

Log adalah sumber informasi berharga untuk mendiagnosis masalah dan melacak aktivitas aplikasi.



Physical Server *vs* Cloud Server

Logging is not limited to the monitoring of physical systems. It can be used to monitor the performance of applications, network traffic, etc.

What is Server Monitoring?

Server monitoring is the process of monitoring a server to ensure that it is operating as expected. Traditionally, this has included CPU usage, or server being up or down. This can include checking for **performance issues**, **checking for errors and suspicious activity**, and monitoring system resources. Server monitoring also includes keeping an *eye* on system health and **diagnosing problems**.

What is Cloud Monitoring?

Whether you are using Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS) or Function as a Service (FaaS) or Serverless, Cloud Monitoring provides visibility into the **performance, uptime, and overall health** of cloud-powered applications. This data can then be visualized on charts and dashboards so that users can identify any **potential issues**.

Bagaimana cara kita agar dapat mengoptimalkan penanganan error dalam program dan meningkatkan kualitas (*quality*) dan keandalan (*reliability*) aplikasi yang dibuat???

Logging System

Use a logging system to record information about *errors* that occur. These logs can contain details about the *error*, including time, location, and other contextual information that can help in the debugging process.

Beberapa praktik yang umum digunakan untuk melacak dan menangani kesalahan (*error*) dalam program.

- 1) Kita dapat menggunakan sistem *logging* untuk mencatat informasi tentang *error* yang terjadi. Log ini dapat berisi detail tentang *error*, termasuk waktu, lokasi, dan informasi kontekstual lainnya yang dapat membantu dalam proses *debugging*.
- 2) Proses *debugging* merupakan kegiatan untuk mengidentifikasi (*identify*), menganalisis (*analyze*) dan memperbaiki kesalahan (*bug*) dalam sebuah program komputer.
- 3) Tujuan dari proses *debugging* yaitu untuk memastikan bahwa program berjalan sesuai dengan yang diharapkan dan menghilangkan masalah yang menyebabkan program tidak berfungsi dengan benar.



Exception Handling

Use the existing **exception handling mechanism** in the programming language used. By using try-catch blocks, we can catch and handle exceptions that may occur in the program.

Beberapa praktik yang umum digunakan untuk melacak dan menangani kesalahan (*error*) dalam program.

- 1) Kita dapat menggunakan mekanisme penanganan *exception* yang ada dalam bahasa pemrograman yang digunakan. Dengan menggunakan *try-catch blok*, kita dapat menangkap dan mengatasi *exception* yang mungkin terjadi dalam program.
- 2) Proses *debugging* merupakan kegiatan untuk mengidentifikasi (*identify*), menganalisis (*analyze*) dan memperbaiki kesalahan (*bug*) dalam sebuah program komputer.
- 3) Tujuan dari proses *debugging* yaitu untuk memastikan bahwa program berjalan sesuai dengan yang diharapkan dan menghilangkan masalah yang menyebabkan program tidak berfungsi dengan benar.



Monitoring & Alerting

Install a monitoring system that can monitor application performance and detect errors. We can use third-party services such as New Relic, Datadog, or Rollbar to monitor the application and receive notifications or alarms when errors occur.

Beberapa praktik yang umum digunakan untuk melacak dan menangani kesalahan (*error*) dalam program.

- 1) Kita dapat memasang **sistem monitoring** yang dapat memantau kinerja aplikasi dan mendeteksi error.
- 2) Proses *debugging* merupakan kegiatan untuk mengidentifikasi (*identify*), menganalisis (*analyze*) dan memperbaiki kesalahan (*bug*) dalam sebuah program komputer.
- 3) Tujuan dari proses *debugging* yaitu untuk memastikan bahwa program berjalan sesuai dengan yang diharapkan dan menghilangkan masalah yang menyebabkan program tidak berfungsi dengan benar.



Unit Testing

Perform thorough *unit testing* to ensure that the code program works properly. By running a series of automated tests, we can detect *errors* early and fix them before the code is widely used.

Beberapa praktik yang umum digunakan untuk melacak dan menangani kesalahan (*error*) dalam program.

- 1) Kita dapat melakukan pengujian unit secara menyeluruh untuk memastikan bahwa kode program berfungsi dengan baik. Dengan menjalankan serangkaian **tes otomatis**, kita dapat mendeteksi error secara dini dan memperbaikinya sebelum kode tersebut digunakan secara luas.
- 2) Proses *debugging* merupakan kegiatan untuk mengidentifikasi (*identify*), menganalisis (*analyze*) dan memperbaiki kesalahan (*bug*) dalam sebuah program komputer.
- 3) Tujuan dari proses *debugging* yaitu untuk memastikan bahwa program berjalan sesuai dengan yang diharapkan dan menghilangkan masalah yang menyebabkan program tidak berfungsi dengan benar.



Version Control

Use a version **control system like Git** to keep track of code changes. This allows us to go back to a previous version if an *error* occurs after certain changes, as well as facilitating collaboration with other development teams.

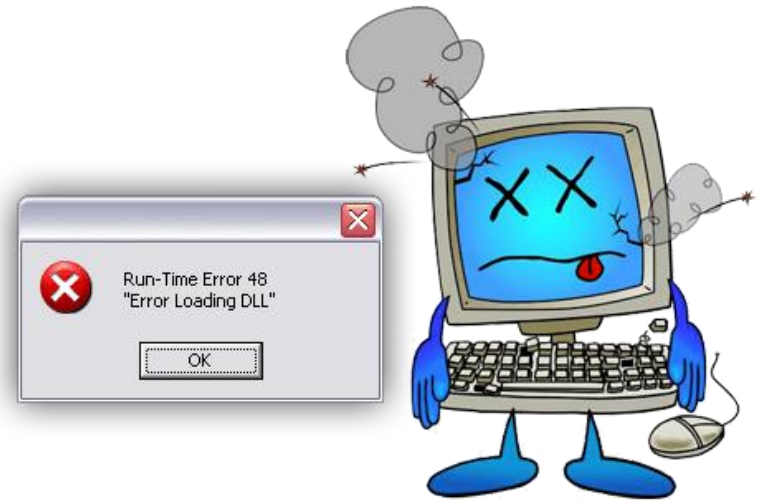
Beberapa praktik yang umum digunakan untuk melacak dan menangani kesalahan (*error*) dalam program.

- 1) Kita dapat menggunakan *sistem kontrol* versi seperti Git untuk melacak perubahan *kode*. Hal ini memungkinkan kita untuk *kembali ke versi sebelumnya jika terjadi error* setelah perubahan tertentu, serta memudahkan kolaborasi dengan tim pengembang lainnya.
- 2) Proses *debugging* merupakan kegiatan untuk mengidentifikasi (*identify*), menganalisis (*analyze*) dan memperbaiki kesalahan (*bug*) dalam sebuah *program komputer*.
- 3) Tujuan dari proses *debugging* yaitu untuk memastikan bahwa program berjalan sesuai dengan yang diharapkan dan menghilangkan masalah yang menyebabkan program tidak berfungsi dengan benar.





Software Bug



Error Code

Dalam konteks pengembangan *software*, apa perbedaan dari *bug* dan *error*?

No Error No Worry

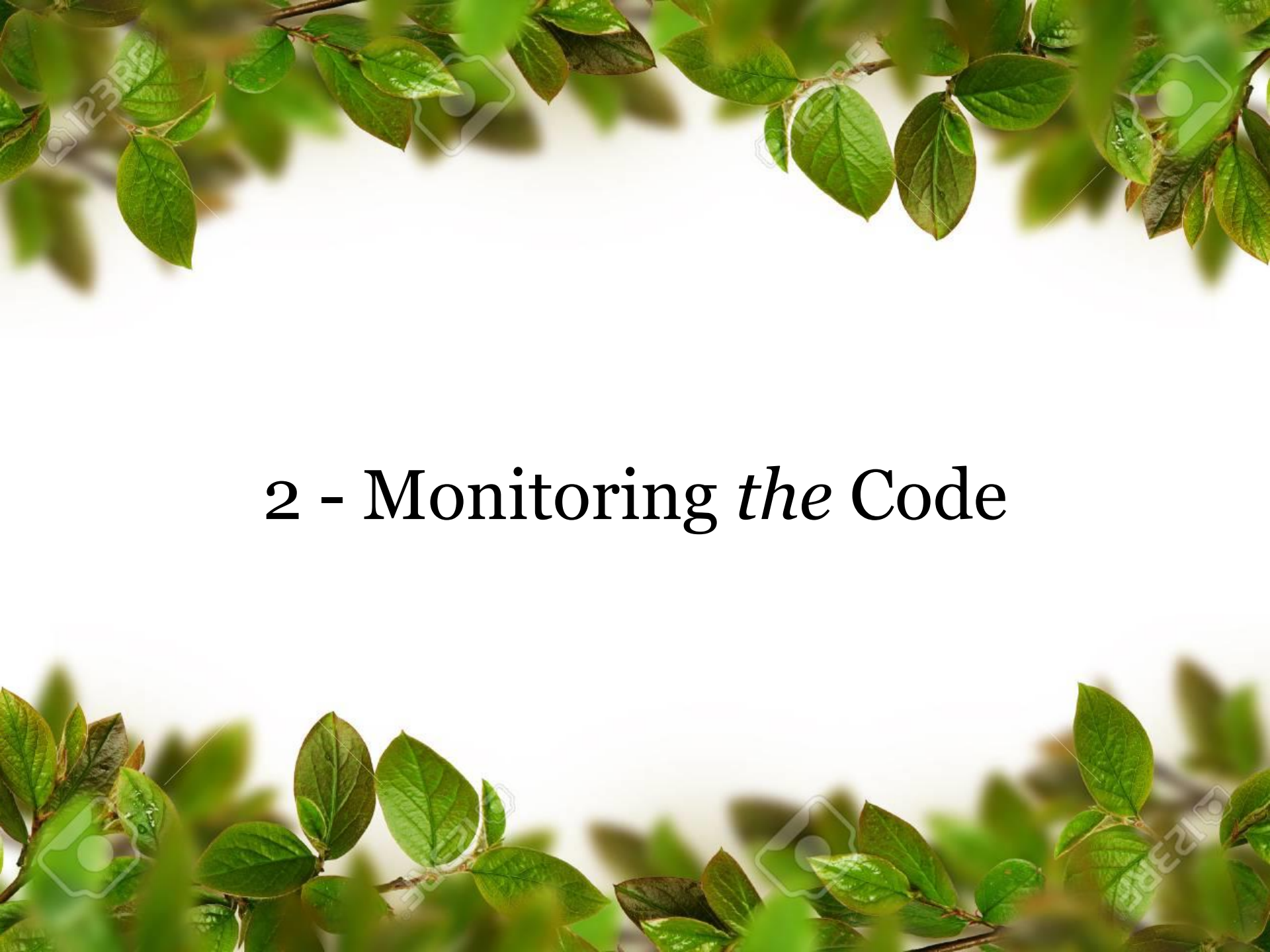


output :

YOU ARE NOT ALONE
YOU ARE NOT ALONE
YOU ARE NOT ALONE

NGK ADA ERROR
NGK SERU !!!

Sistem yang kompleks seringkali rentan terhadap **situasi tak terduga** atau perubahan lingkungan dan kadang-kadang bahkan **kesalahan kecil** dapat memiliki **dampak besar**.

The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green with some visible veins and slight shadows, creating a natural border around the central text.

2 - Monitoring *the* Code

Chapter 8: Monitoring the Code

Nagios

Munin

Ganglia

Graphite

Log handling

Client-side logging libraries

The ELK stack

Summary

Chapter 9: Issue Tracking

What are issue trackers used for?

Some examples of workflows and issues

What do we need from an issue tracker?

Problems with issue tracker proliferation

All the trackers

Bugzilla

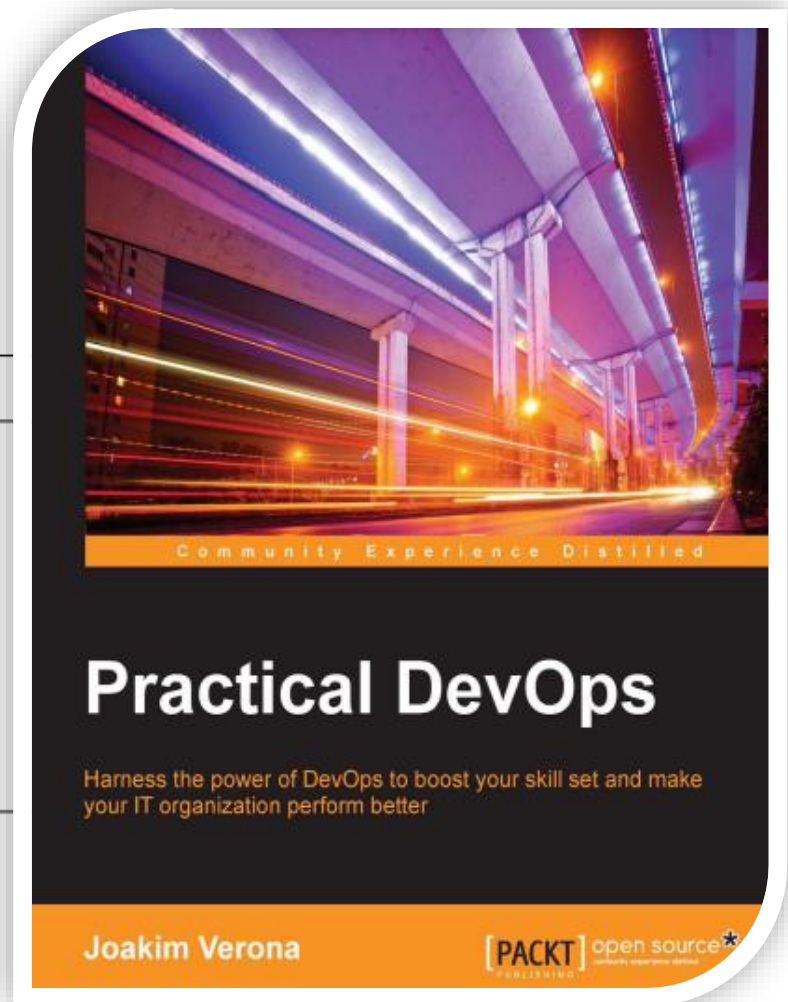
Trac

Redmine

The GitLab issue tracker

Jira

Summary



167
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8

Monitoring the Code

In the previous chapter, we explored methods of deploying our code.

Now that the code has been safely deployed to your servers with the deployment solution of your choice, you need to watch over it to make sure it's running properly. You can spend a lot of time preparing for the many modes of failure that you may envision during development. In the end, your software will probably break for other reasons altogether than those you prepared for. If your system breaks, it can be very expensive for your organization, either in lost revenue or in terms of lost credibility, which, in the end, might amount to the same thing. You need to know as quickly as possible what has gone wrong in order to deal with the situation.

Given the potential negative impact of service downtime, there are many alternative solutions that approach the problem domain of watching over your deployed code from different angles and viewpoints.

In this chapter, we will have a look at several of the options that are available to us.

Now that the code has been safely deployed to your servers with the deployment solution of your choice, you need to watch over it to make sure it's running properly. You can spend a lot of time preparing for the many modes of failure that you may envision during development.

Developing Efficient Server Monitoring Systems Using AI for Real-time Data Processing

International Journal of Engineering and Technology Research (IJETR), Volume 8, Issue 1, January-December 2023, pp. 26-37

12 Pages • Posted: 22 Jan 2025

Sagar Vishnubhai Sheta

Independent

Date Written: August 26, 2023

Abstract

This paper explores the development and implementation of AI-based server monitoring systems aimed at optimizing real-time data processing. Traditional server monitoring systems, reliant on static thresholds and manual oversight, often struggle to handle the scale and complexity of modern server environments. In contrast, AI-driven monitoring leverages machine learning and deep learning algorithms to analyze large volumes of data, enabling precise anomaly detection and predictive maintenance. The proposed AI-based system architecture incorporates components such as real-time data collection, preprocessing, and model deployment, which together enhance response times, improve anomaly detection accuracy, and reduce system downtime. This study discusses the comparative advantages of AI-based monitoring over traditional methods, highlights performance and efficiency gains, and addresses key considerations for scalability and adaptability in diverse server environments. The findings underscore AI's role in advancing server monitoring from reactive to proactive management, providing a foundation for future research in autonomous monitoring solutions.

Real-Time Diagnosis of Configuration Errors for Software of AI Server Infrastructure

Publisher: IEEE

Cite This

PDF

Guangquan Xu  ; Xinru Ding  ; Sihan Xu  ; Yan Jia ; Shaoying Liu  ; Shicheng Feng [All Authors](#)

209
Full
Text Views



- Abstract
- Authors
- Keywords
- Metrics

Abstract:

Artificial intelligence (AI) server infrastructure has been built to support AI applications and handle data-intensive workloads. AI server infrastructure is the essential building blocks, and errors in AI server infrastructure may lead to fatal consequences to any AI applications built upon it. Compared to traditional software, software for AI server infrastructure is more configurable, and thus more likely to have configuration errors that might prevent correct software behaviors. Previous work on misconfiguration diagnosis requires sufficient execution history or manual intervention, and can hardly diagnose potential misconfigurations which are not triggered at launching. In this paper, we propose a real-time method to address these issues. Specifically, we combine program analysis and real-time log parsing to diagnose configuration errors. It maps each configuration option to the log code by applying program slicing only once, and parses real-time logs during the operation of the AI server without manual intervention. We evaluate the effectiveness of our



Using a multi-agent system and artificial intelligence for monitoring and improving the cloud performance and security

Daniel Grzonka ^a  , Agnieszka Jakóbik ^a , Joanna Kołodziej ^b , Sabri Pllana ^c 

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<https://doi.org/10.1016/j.future.2017.05.046> 

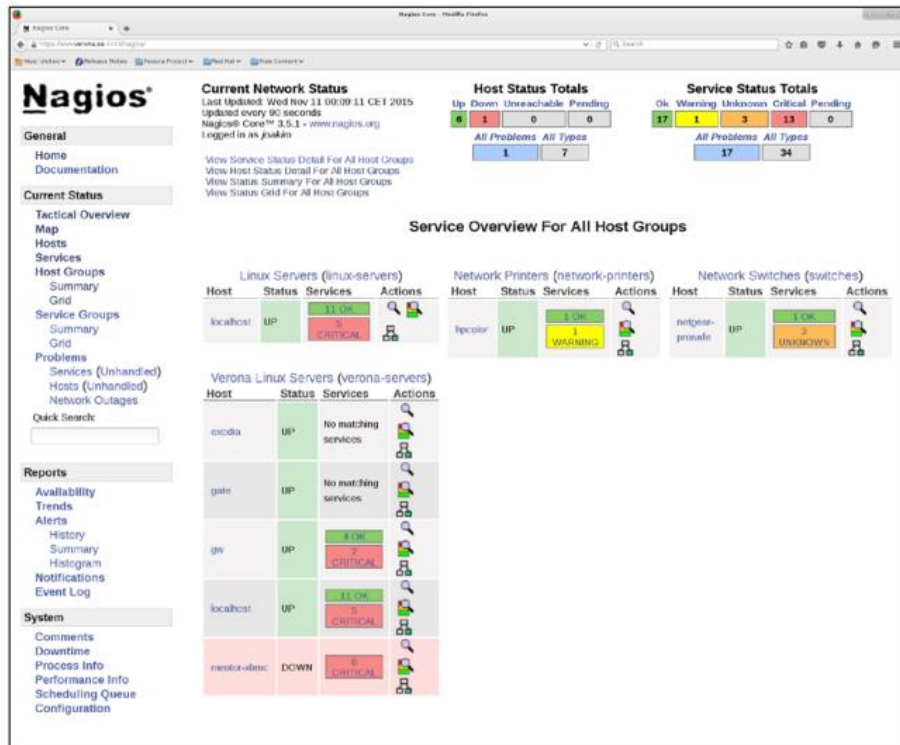
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Highlights

- We propose the new Multi-Agent based Cloud Monitoring (MAS-CM) model.
- Our system monitors tasks gathering, scheduling and execution processes.
- MAS-CM model is designed to control the performance and chosen security objectives.
- We focus on prevention of tasks injection and optimization of scheduling process.
- The effectiveness of MAS-CM has been proofed by an empirical analysis.

Monitoring Solution *for* Server Health

Nagios has been around since 1999 and has a large community that provides plugins and extensions for various needs. There are extensive resources available for Nagios on the web, and commercial support if your organization requires it. Since it has been around so long and many organizations use it, **Nagios is the standard in network monitoring against which other solutions compare themselves**. As such, it is the natural starting point for our journey through the monitoring landscape.



Munin

Munin is used to graph server statistics such as memory usage, which is useful in order to understand overall server health. Since Munin graphs statistics over time, you can see resource allocation trends, which can help you find problems before they get serious. There are several other applications like Munin that create graphs; however, like Nagios, Munin is a good starting point.

It is designed to be easy to use and set up. The out-of-the-box experience gives you many graphs with little work.

In the legends of the Norse, Hugin and Munin were two pitch-black ravens. They tirelessly flew around the world of Midgard and collected information. After their journeys far and wide, they returned to the god king Odin to sit on his shoulders and tell him all about the world. Hugin was the raven of thought and Munin of memory.

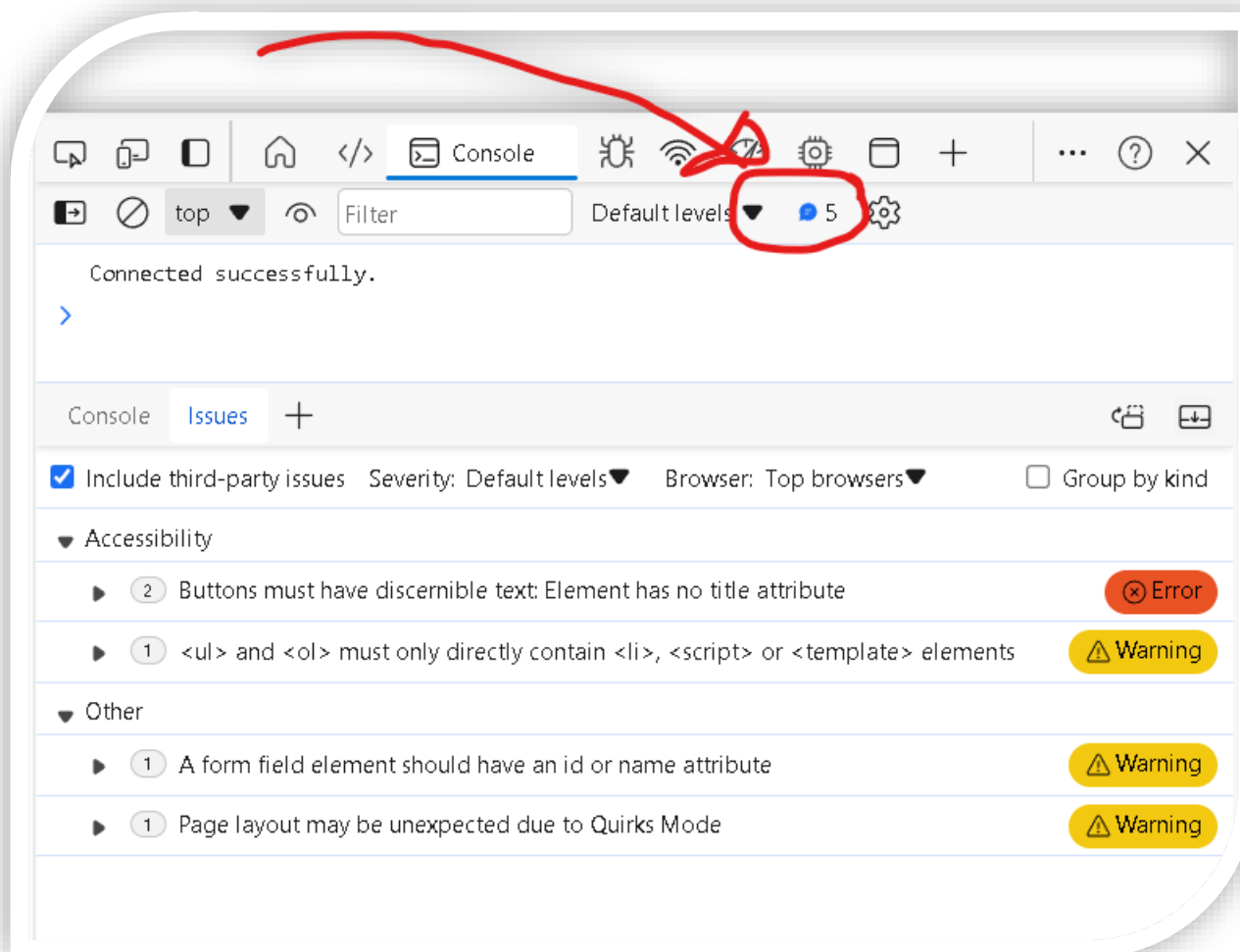
Ganglia

Ganglia is a graphing and monitoring solution for large clusters. It can aggregate information in convenient overview displays.

The word "ganglia" is the plural form of ganglion, which is a nerve cell cluster in anatomy. The analogy implies that Ganglia can be the sensory nerve cell network in your cluster.

Like Munin, Ganglia also uses RRDs for database storage and graphing, so the graphs will look similar to the previous Munin graphs. Code reuse is a good thing!

Ganglia has an interesting online demonstration at <http://ganglia.wikimedia.org/latest/>.



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3 - Monitoring Log *in* Cloud Server

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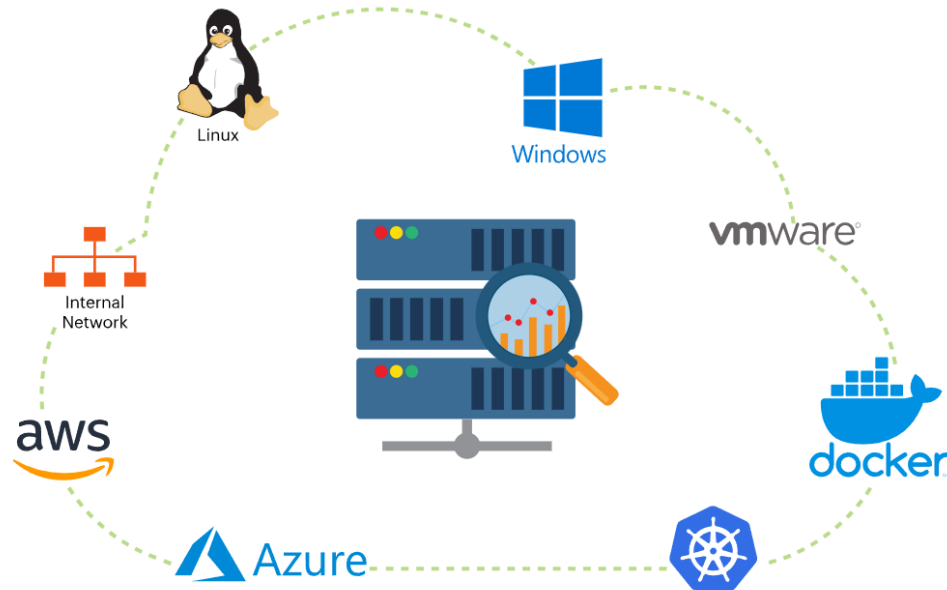
Monitoring Log

We can **monitor application logs** regularly and continuously. Monitoring these logs can help us identify error patterns, understand the underlying causes, and take appropriate preventive actions.



Kita dapat melakukan monitoring terhadap log aplikasi secara berkala dan terus-menerus. Pemantauan log ini dapat membantu kita **mengidentifikasi pola error**, **memahami penyebab yang mendasarinya**, dan **mengambil tindakan pencegahan** yang tepat.

Cloud Server Monitoring



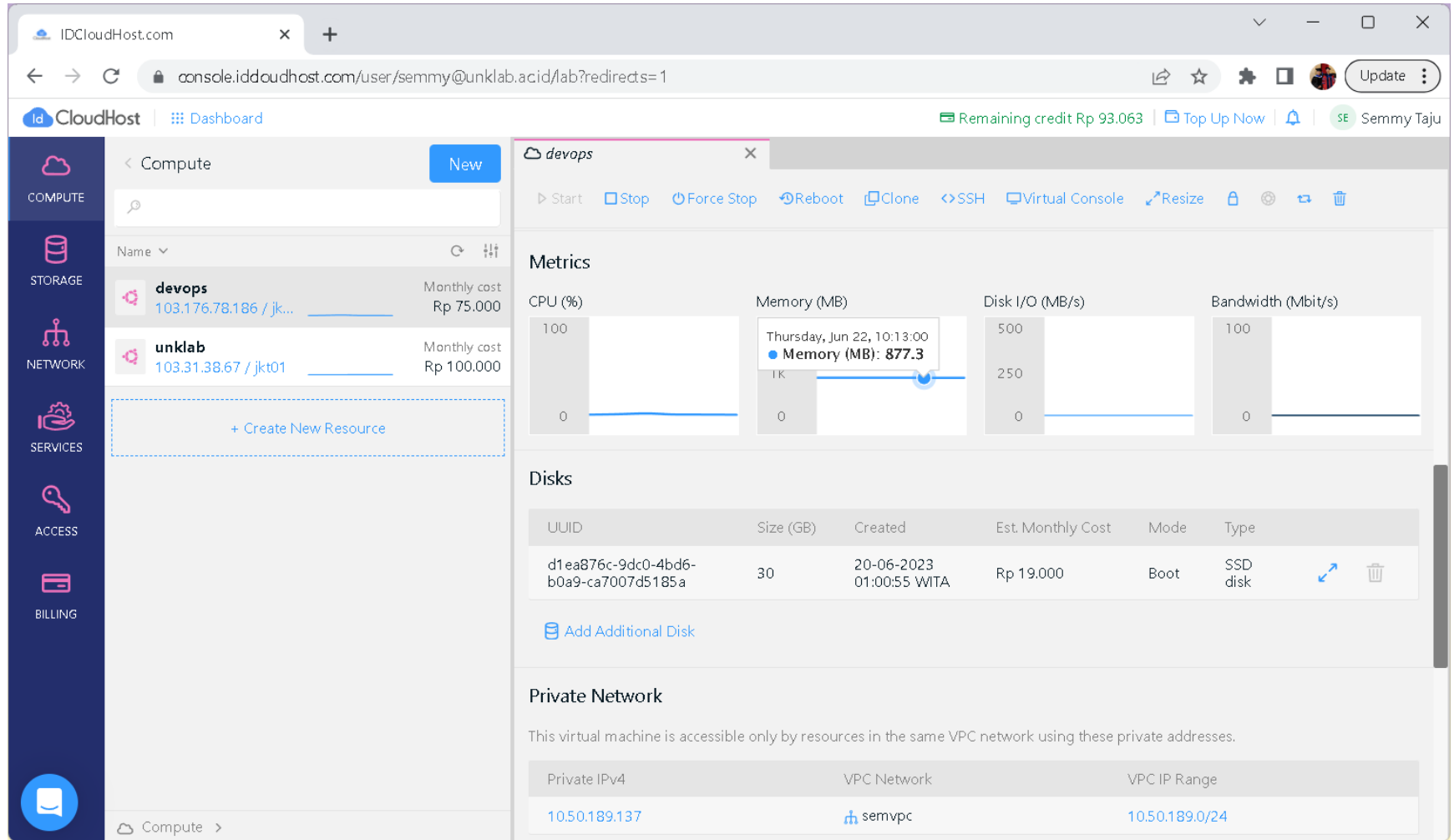
Monitoring di *cloud* seringkali dapat menjadi lebih **hemat biaya** dibandingkan dengan pemantauan pada **server fisik** **atau infrastruktur lokal**. **Cloud service** memberikan kemampuan untuk mengatur dan menyesuaikan sumber daya pemantauan sesuai dengan kebutuhan.

Cloud Server Monitoring



- ❖ **Cloud monitoring** provides more general information than server monitoring, while server monitoring is more focused on specific aspects of server **health** and **performance**.
- ❖ **Cloud logging** provides detailed information about all activity taking place in your cloud, including *errors*, *events*, and *user interactions*.
- ❖ This information can be used to monitor and **troubleshoot issues** as they happen, rather than waiting for reports or alerts to come in.

Real-time Cloud Server Monitoring

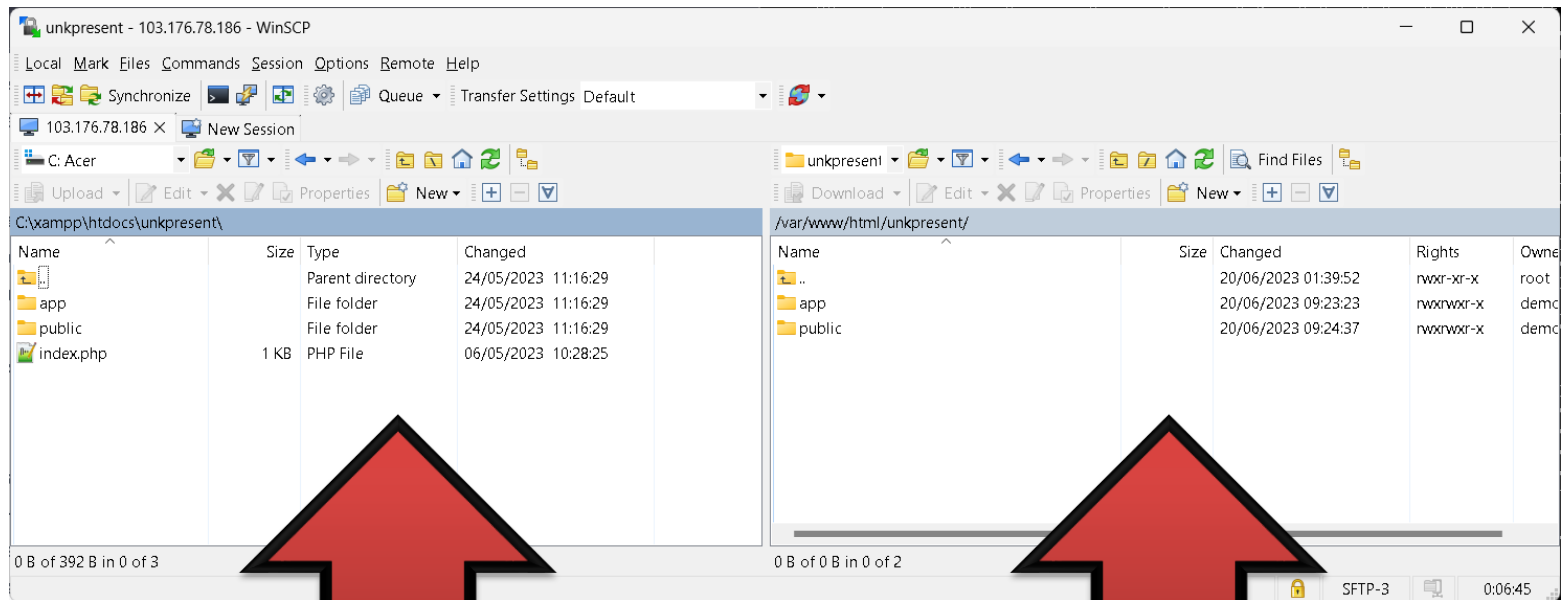




Exercise *for* Students

WinSCP (Windows Secure Copy)

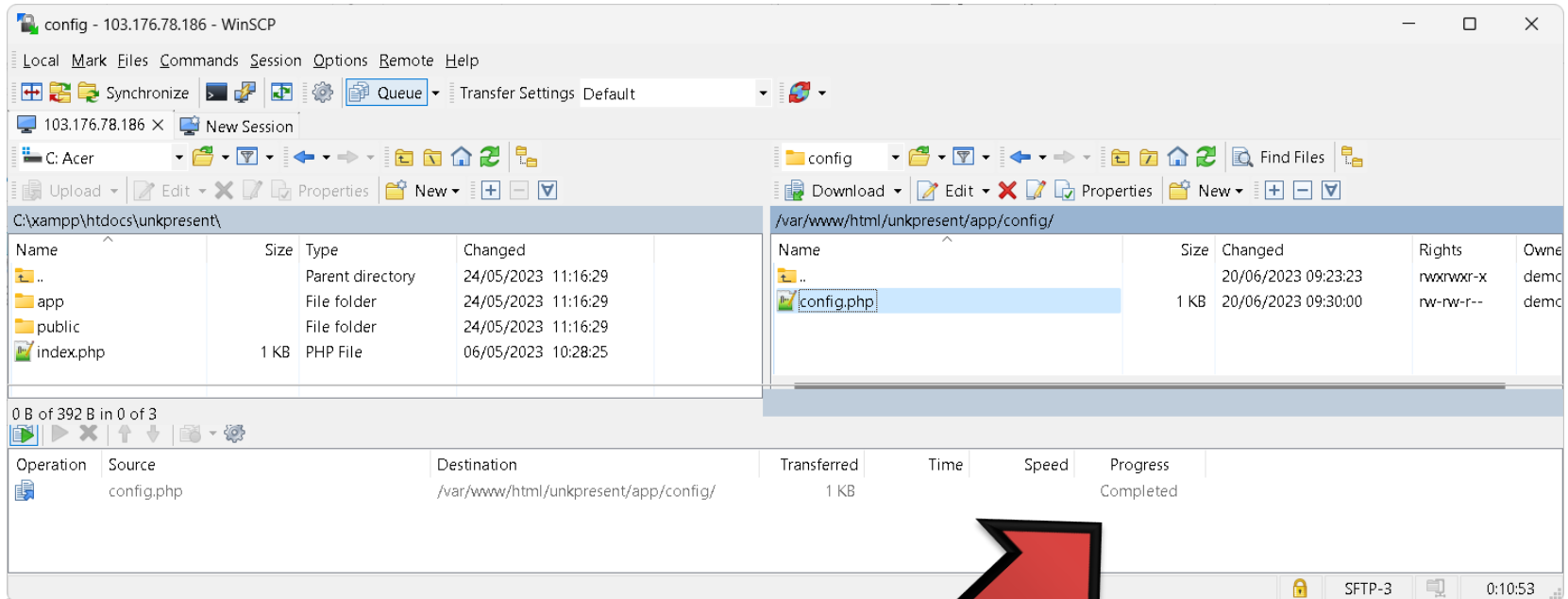
WinSCP (Windows Secure Copy) menyediakan interface grafis untuk mentransfer file antara komputer lokal dan *cloud server* melalui protokol Secure Copy (SCP) atau File Transfer Protocol Secure (SFTP).



Local Computer

Cloud Server

Check Transfer File Progress



Mengecek transfer file progress di WinSCP sangat penting karena memberikan informasi tentang status dan kemajuan proses transfer file yang sedang berlangsung antara komputer lokal dan cloud server.

Command Tools #1

Monitoring Performance Ubuntu Server 22.04

Monitoring server adalah kegiatan memantau sumber daya sistem server seperti penggunaan CPU (CPU Usage), konsumsi memori (Memory RAM Consumption), jaringan, penggunaan disk (disk usage), dan lain sebagainya.

Ubuntu Command Top

Command Top berfungsi untuk menampilkan kinerja sistem secara real time seperti CPU, Memory, Serta proses yang sedang berjalan.

Ketikkan Command dibawah ini Terminal:

> **top**
> **Ctrl+c (exit)**

```
demouser@devops: ~  
top - 01:14:15 up 2 days, 8:12, 3 users, load average: 0.00, 0.00, 0.00  
Tasks: 124 total, 1 running, 123 sleeping, 0 stopped, 0 zombie  
%Cpu(s): 0.3 us, 0.3 sy, 0.0 ni, 98.3 id, 1.0 wa, 0.0 hi, 0.0 si, 0.0 st  
MiB Mem : 971.0 total, 98.2 free, 638.8 used, 233.9 buff/cache  
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 184.5 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
4727	mysql	20	0	1819364	409504	10216	S	1.0	41.2	35:18.01	mysqld
15820	root	20	0	78180	21064	4128	S	0.3	2.1	5:15.96	apt
42870	demouser	20	0	17296	7940	5524	S	0.3	0.8	0:00.01	sshd
42883	demouser	20	0	10900	3704	3136	R	0.3	0.4	0:00.02	top
1	root	20	0	167512	10464	5676	S	0.0	1.1	0:23.63	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.04	kthreadd
3	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_gp
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	rcu_par_gp
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_highpri
9	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	mm_percpu_wq
10	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_tasks_rude_
11	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_tasks_trace
12	root	20	0	0	0	0	S	0.0	0.0	0:00.42	ksoftirqd/0
13	root	20	0	0	0	0	I	0.0	0.0	0:05.04	rcu_sched
14	root	rt	0	0	0	0	S	0.0	0.0	0:01.40	migration/0
15	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
16	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
17	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
18	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/1

Ubuntu Command Htop

Command Htop merupakan perintah untuk memonitoring proses yang sedang berjalan pada sistem melalui terminal linux. Sama seperti perintah Top, Htop berfungsi untuk memantau penggunaan memori, total swap yang digunakan, total CPU yang sedang berjalan, memantau aplikasi apa saja yang sedang berjalan, dan seberapa lama komputer sudah berjalan.

Ketikkan Command dibawah ini Terminal:

> **htop**

> **Ctrl+c OR F10 (exit)**

```
demouser@devops: ~  
  
 0.7%] Tasks: 52, 74 thr; 1 running  
 0.0%] Load average: 0.12 0.04 0.01  
Mem[|||||642M/971M] Uptime: 2 days, 08:16:02  
Swp[|||||0K/0K]  
  
 PID USER      PRI  NI  VIRT   RES   SHR  S  CPU% MEM%   TIME+  Command  
 4727 mysql     20    0 1776M  399M 10216 S  1.3 41.2 35:20.06 /usr/sbin/mysqld  
 4745 mysql     20    0 1776M  399M 10216 S  0.7 41.2 24:57.15 /usr/sbin/mysqld  
 42900 demouser 20    0  8436  4064  3424 R  0.7  0.4  0:00.13 htop  
    1 root      20    0  163M 10464  5676 S  0.0  1.1  0:23.63 /sbin/init  
 344 root      19   -1 64236 11152  9884 S  0.0  1.1  0:27.72 /lib/systemd/systemd-journald  
 382 root      RT    0  282M 27104  9076 S  0.0  2.7  0:28.94 /sbin/multipathd -d -s  
 385 root      20    0 11848  4980  2796 S  0.0  0.5  0:01.18 /lib/systemd/systemd-udevd  
 386 root      20    0  282M 27104  9076 S  0.0  2.7  0:00.00 /sbin/multipathd -d -s  
 387 root      RT    0  282M 27104  9076 S  0.0  2.7  0:00.00 /sbin/multipathd -d -s  
 388 root      RT    0  282M 27104  9076 S  0.0  2.7  0:00.00 /sbin/multipathd -d -s  
 389 root      RT    0  282M 27104  9076 S  0.0  2.7  0:00.42 /sbin/multipathd -d -s  
 390 root      RT    0  282M 27104  9076 S  0.0  2.7  0:20.58 /sbin/multipathd -d -s  
 391 root      RT    0  282M 27104  9076 S  0.0  2.7  0:00.00 /sbin/multipathd -d -s  
 475 systemd-t 20    0 89300  3764  2960 S  0.0  0.4  0:00.64 /lib/systemd/systemd-timesyncd  
 486 systemd-t 20    0 89300  3764  2960 S  0.0  0.4  0:00.00 /lib/systemd/systemd-timesyncd  
 527 systemd-r 20    0 25348  8452  4312 S  0.0  0.9  0:00.64 /lib/systemd/systemd-resolved  
 588 systemd-n 20    0 16068  4296  3260 S  0.0  0.4  0:01.21 /lib/systemd/systemd-networkd  
 679 root      20    0  7284  2536  2280 S  0.0  0.3  0:00.59 /usr/sbin/cron -f -P  
  
F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice F8Nice F9Kill F10Quit
```

VmStat (Virtual Memory Statistics)

VmStat merupakan perintah pada Linux yang berfungsi untuk menampilkan penggunaan memori, swap, memberikan informasi interrupt system, kecepatan I/O, statistik CPU secara real time, dan proses apa saja yang sedang berjalan.

Ketikkan Command dibawah ini Terminal:

```
> vmstat
```

[illegible]

Command Tools #2

Log Analysis & Attack Detection on Ubuntu Server (Apache2)

Ubuntu Server 22.04

Apache is part of the **LAMP stack** of software for **Linux (Linux, Apache, MySQL, PHP)**. Apache is responsible for serving web pages to people looking at our website.

Focus file log server, seperti:

- /var/log/apache2/access.log (aktivitas web server)
- /var/log/auth.log (aktivitas login SSH)
- /var/log/apache2/error.log (error aplikasi/server)

Install GoAccess in Ubuntu 22.04

GoAccess is an open source real-time web log analyzer and interactive viewer that runs in a terminal in unix systems or through your browser.

Install dan cek versi dari repositori Ubuntu

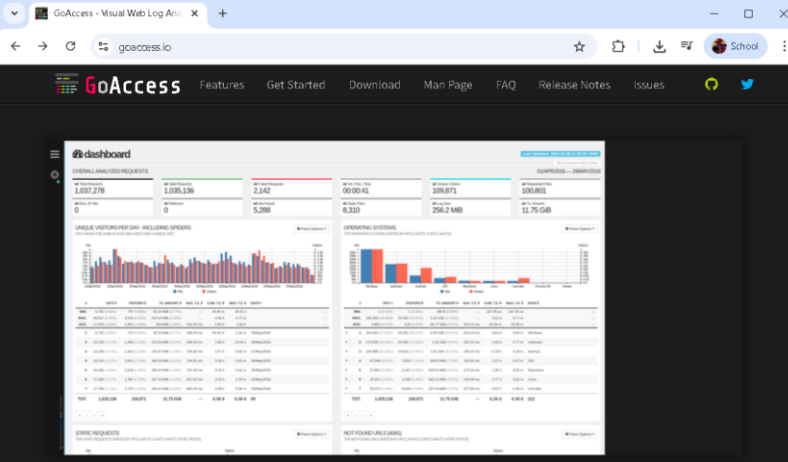
- `sudo apt install goaccess -y`
- `goaccess -version`
- `goaccess -V`

```
root@PresensiApp2025: /home/presensi2025admin# goaccess -V
No VM guests are running outdated hypervisor (qemu) binaries on this host.
GoAccess - 1.5.5.
For more details visit: https://goaccess.io/
Copyright (C) 2009-2022 by Gerardo Orellana

Build configure arguments:
--enable-utf8
--enable-geoip=mmdb
--with-openssl
root@PresensiApp2025: /home/presensi2025admin# goaccess --version
GoAccess - 1.5.5.
For more details visit: https://goaccess.io/
Copyright (C) 2009-2022 by Gerardo Orellana

Build configure arguments:
--enable-utf8
--enable-geoip=mmdb
--with-openssl
root@PresensiApp2025: /home/presensi2025admin# |
```

<https://goaccess.io/>



The screenshot shows the GoAccess web interface in a browser. The dashboard includes several key metrics at the top: Total Requests (1,000,000), Unique Visitors (100,000), and Average Session Length (1:30). Below these are two bar charts: 'UNIQUE VISITORS PER DAY' and 'OPERATING SYSTEMS'. The bottom section contains a table of 'TOP REQUESTS' and a 'NOT FOUND REQUESTS' section.

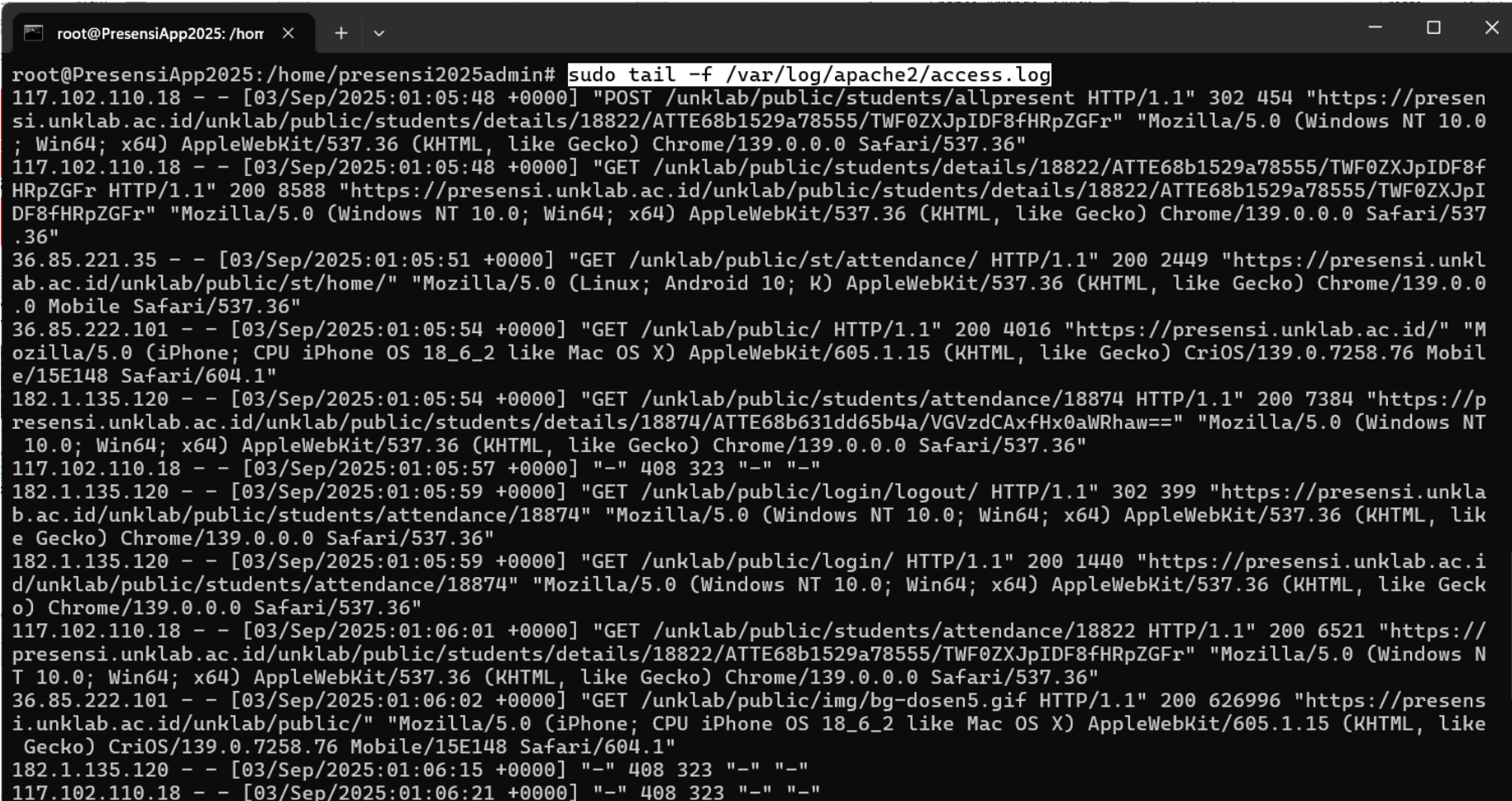
What is it?

GoAccess is an open source **real-time web log analyzer** and interactive viewer that runs in a **terminal** in ***nix** systems or through your **browser**.

Cek lokasi log Apache

Default log Apache ada di: /var/log/apache2/access.

➤ *sudo tail -f /var/log/apache2/access.log*



```
root@PresensiApp2025: /horr x + v
root@PresensiApp2025:/home/presensi2025admin# sudo tail -f /var/log/apache2/access.log
117.102.110.18 - - [03/Sep/2025:01:05:48 +0000] "POST /unklab/public/students/allpresent HTTP/1.1" 302 454 "https://presensi.unklab.ac.id/unklab/public/students/details/18822/ATTE68b1529a78555/TWF0ZXJpIDF8fHRpZGFr" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/139.0.0.0 Safari/537.36"
117.102.110.18 - - [03/Sep/2025:01:05:48 +0000] "GET /unklab/public/students/details/18822/ATTE68b1529a78555/TWF0ZXJpIDF8fHRpZGFr HTTP/1.1" 200 8588 "https://presensi.unklab.ac.id/unklab/public/students/details/18822/ATTE68b1529a78555/TWF0ZXJpIDF8fHRpZGFr" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/139.0.0.0 Safari/537.36"
36.85.221.35 - - [03/Sep/2025:01:05:51 +0000] "GET /unklab/public/st/attendance/ HTTP/1.1" 200 2449 "https://presensi.unklab.ac.id/unklab/public/st/home/" "Mozilla/5.0 (Linux; Android 10; K) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/139.0.0.0 Mobile Safari/537.36"
36.85.222.101 - - [03/Sep/2025:01:05:54 +0000] "GET /unklab/public/ HTTP/1.1" 200 4016 "https://presensi.unklab.ac.id/" "Mozilla/5.0 (iPhone; CPU iPhone OS 18_6_2 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) CriOS/139.0.7258.76 Mobile/15E148 Safari/604.1"
182.1.135.120 - - [03/Sep/2025:01:05:54 +0000] "GET /unklab/public/students/attendance/18874 HTTP/1.1" 200 7384 "https://presensi.unklab.ac.id/unklab/public/students/details/18874/ATTE68b631dd65b4a/VGVzdCAxfHx0aWRhaw==" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/139.0.0.0 Safari/537.36"
117.102.110.18 - - [03/Sep/2025:01:05:57 +0000] "-" 408 323 "-" "-"
182.1.135.120 - - [03/Sep/2025:01:05:59 +0000] "GET /unklab/public/login/logout/ HTTP/1.1" 302 399 "https://presensi.unklab.ac.id/unklab/public/students/attendance/18874" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/139.0.0.0 Safari/537.36"
182.1.135.120 - - [03/Sep/2025:01:05:59 +0000] "GET /unklab/public/login/ HTTP/1.1" 200 1440 "https://presensi.unklab.ac.id/unklab/public/students/attendance/18874" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/139.0.0.0 Safari/537.36"
117.102.110.18 - - [03/Sep/2025:01:06:01 +0000] "GET /unklab/public/students/attendance/18822 HTTP/1.1" 200 6521 "https://presensi.unklab.ac.id/unklab/public/students/details/18822/ATTE68b1529a78555/TWF0ZXJpIDF8fHRpZGFr" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/139.0.0.0 Safari/537.36"
36.85.222.101 - - [03/Sep/2025:01:06:02 +0000] "GET /unklab/public/img/bg-dosen5.gif HTTP/1.1" 200 626996 "https://presensi.unklab.ac.id/unklab/public/" "Mozilla/5.0 (iPhone; CPU iPhone OS 18_6_2 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) CriOS/139.0.7258.76 Mobile/15E148 Safari/604.1"
182.1.135.120 - - [03/Sep/2025:01:06:15 +0000] "-" 408 323 "-" "-"
117.102.110.18 - - [03/Sep/2025:01:06:21 +0000] "-" 408 323 "-" "-"
```

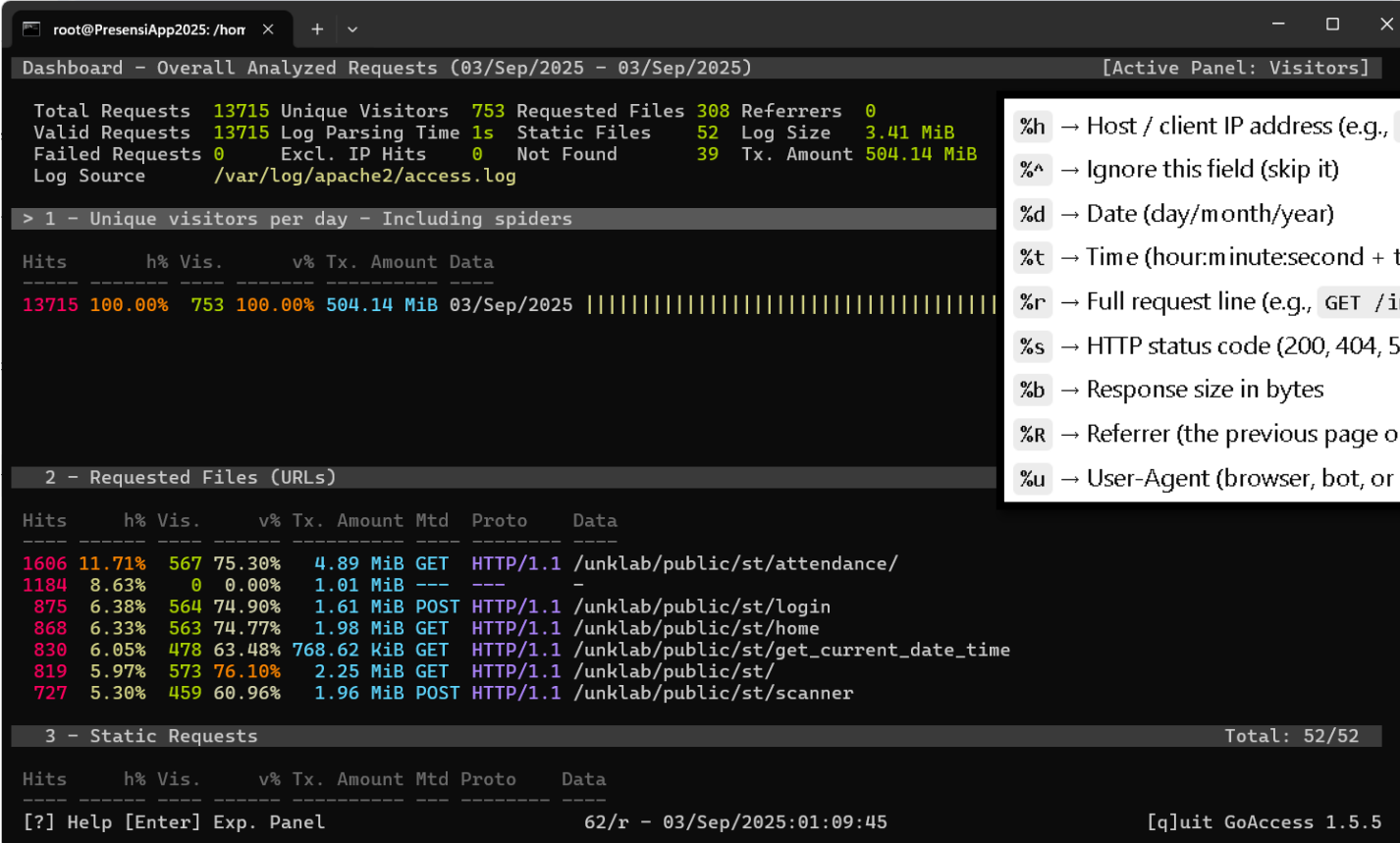
Jalankan GoAccess di Terminal

Jalankan GoAccess di Terminal:

➤ `sudo goaccess /var/log/apache2/access.log --log-format=COMBINED`

Format COMBINED sesuai dengan default Apache log format:

`%h %^[%d:%t %^] "%r" %s %b "%R" "%u"`



root@PresensiApp2025: /horr x + v

Dashboard - Overall Analyzed Requests (03/Sep/2025 - 03/Sep/2025) [Active Panel: Visitors]

Total Requests	13715	Unique Visitors	753	Requested Files	308	Referrers	0
Valid Requests	13715	Log Parsing Time	1s	Static Files	52	Log Size	3.41 MiB
Failed Requests	0	Excl. IP Hits	0	Not Found	39	Tx. Amount	504.14 MiB
Log Source	/var/log/apache2/access.log						

> 1 - Unique visitors per day - Including spiders

Hits	h% Vis.	v% Tx. Amount	Data
13715	100.00%	753 100.00%	504.14 MiB 03/Sep/2025

2 - Requested Files (URLs)

Hits	h% Vis.	v% Tx. Amount	Mtd	Proto	Data
1606	11.71%	567 75.30%	4.89 MiB	GET	HTTP/1.1 /unklab/public/st/attendance/
1184	8.63%	0 0.00%	1.01 MiB	---	---
875	6.38%	564 74.90%	1.61 MiB	POST	HTTP/1.1 /unklab/public/st/login
868	6.33%	563 74.77%	1.98 MiB	GET	HTTP/1.1 /unklab/public/st/home
830	6.05%	478 63.48%	768.62 KiB	GET	HTTP/1.1 /unklab/public/st/get_current_date_time
819	5.97%	573 76.10%	2.25 MiB	GET	HTTP/1.1 /unklab/public/st/
727	5.30%	459 60.96%	1.96 MiB	POST	HTTP/1.1 /unklab/public/st/scanner

3 - Static Requests Total: 52/52

Hits	h% Vis.	v% Tx. Amount	Mtd	Proto	Data
------	---------	---------------	-----	-------	------

[?] Help [Enter] Exp. Panel 62/r - 03/Sep/2025:01:09:45 [q]uit GoAccess 1.5.5

%h → Host / client IP address (e.g., 192.168.1.10)

%^ → Ignore this field (skip it)

%d → Date (day/month/year)

%t → Time (hour:minute:second + timezone)

%r → Full request line (e.g., GET /index.php?id=1 HTTP/1.1)

%s → HTTP status code (200, 404, 500, etc.)

%b → Response size in bytes

%R → Referrer (the previous page or source, like Google or another site)

%u → User-Agent (browser, bot, or attack tool)

Dashboard Web *for* GoAccess

GoAccess bisa menampilkan laporan real-time dalam bentuk HTML.

[1] Generate report statis (HTML biasa)

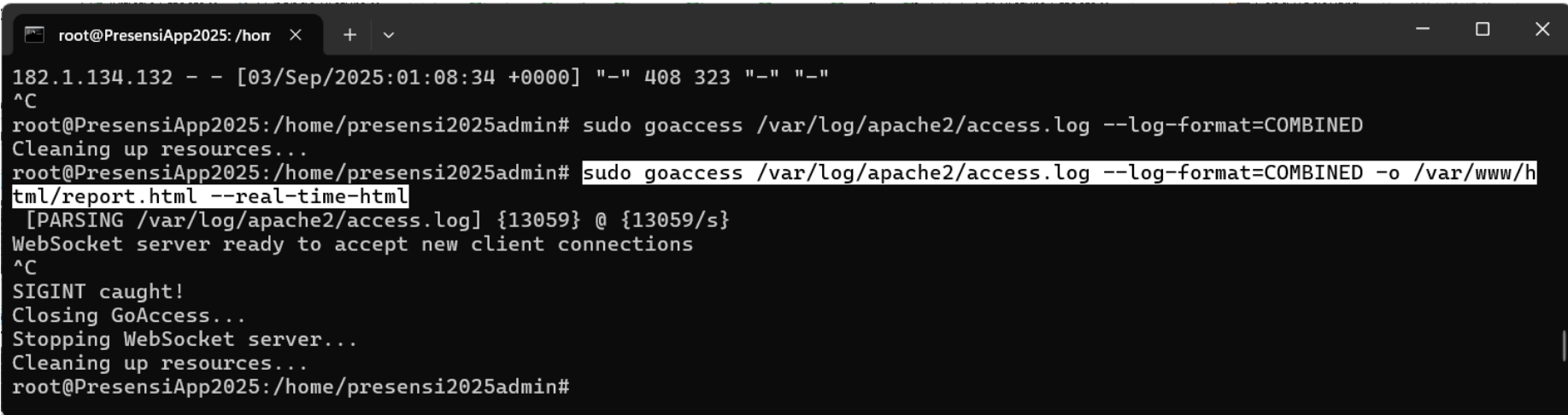
➤ `sudo goaccess /var/log/apache2/access.log --log-format=COMBINED -o /var/www/html/laporan.html`

[2] Generate report real-time (pakai WebSocket server GoAccess)

➤ `sudo goaccess /var/log/apache2/access.log --log-format=COMBINED -o /var/www/html/laporan.html --real-time-html`

[3] Lalu buka di browser:

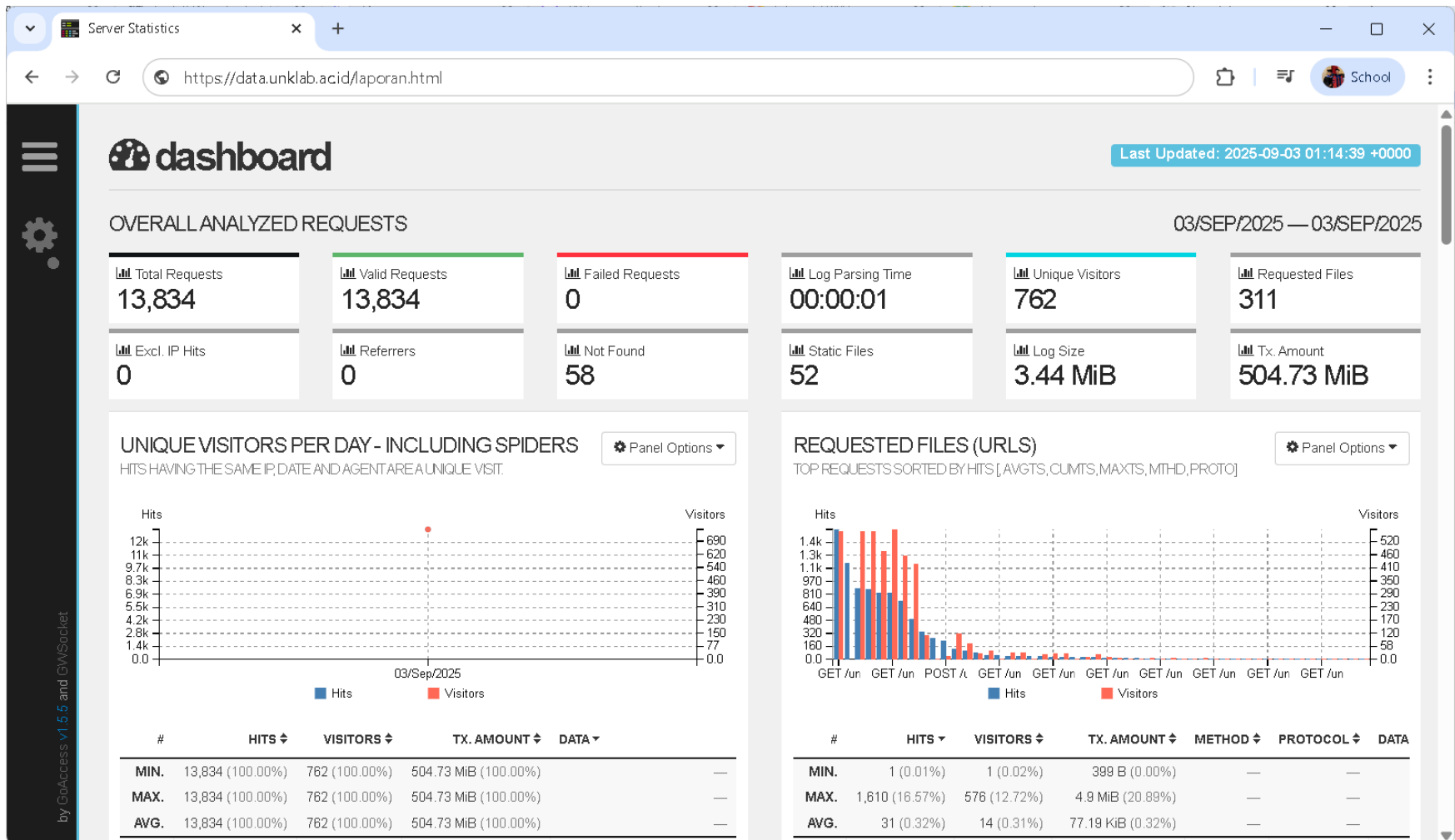
`http://server_ip/report.html`



```
root@PresensiApp2025: /hor  x  +  v
182.1.134.132 - - [03/Sep/2025:01:08:34 +0000] "-" 408 323 "-" "-"
^C
root@PresensiApp2025: /home/presensi2025admin# sudo goaccess /var/log/apache2/access.log --log-format=COMBINED
Cleaning up resources...
root@PresensiApp2025: /home/presensi2025admin# sudo goaccess /var/log/apache2/access.log --log-format=COMBINED -o /var/www/h
tml/report.html --real-time-html
[PARSING /var/log/apache2/access.log] {13059} @ {13059/s}
WebSocket server ready to accept new client connections
^C
SIGINT caught!
Closing GoAccess...
Stopping WebSocket server...
Cleaning up resources...
root@PresensiApp2025: /home/presensi2025admin#
```

Dashboard Web for GoAccess

GoAccess bisa menampilkan laporan real-time dalam bentuk HTML.



Run GoAccess sebagai Service

[1] Buat service supaya GoAccess selalu jalan di background:

➤ *sudo nano /etc/systemd/system/goaccess.service*

[2] Open file dan copy text di bawah ini dan Isi dengan:

[Unit]

Description=GoAccess Real-Time Log Analyzer

After=network.target

[Service]

ExecStart=/usr/bin/goaccess /var/log/apache2/access.log --log-format=COMBINED -o /var/www/html/report.html --real-time-html

Restart=always

[Install]

WantedBy=multi-user.target

[3] Simpan, lalu jalankan:

➤ *sudo systemctl daemon-reload*

➤ *sudo systemctl enable goaccess*

➤ *sudo systemctl start goaccess*

[4] Buka di browser:

http://server_ip/report.html

Analysis Tasks

Log Analysis and Attack Detection on Ubuntu Server (Apache2).

Mahasiswa dapat:

- 1) Menggunakan GoAccess untuk menganalisis log Apache.
- 2) Menemukan IP address dengan request terbanyak.
- 3) Menentukan URL yang paling sering diakses.
- 4) Mengklasifikasikan HTTP status code (200, 404, 500, dll).
- 5) Menganalisis User-Agent (browser, bot, scanner).
- 6) Mengidentifikasi potensi serangan (SQLi, XSS, scanning).

Threat Hunting *dari* Log Server

Mahasiswa diajak **langsung praktik** mencari tanda-tanda serangan (*threat*) di server dengan cara **menganalisis log** apache (catatan aktivitas server).

Task 1	Analisis IP terbanyak request. Catat 5 IP dengan request terbanyak. Apakah ada IP yang terlihat melakukan request berulang dalam jumlah tidak wajar?
Task 2	URL paling sering diakses. Dari output GoAccess (menu Requested URLs), catat 5 URL teratas. Apakah ada URL mencurigakan (misalnya mencoba akses file admin, phpmyadmin, atau wp-login.php padahal bukan WordPress)?
Task 3	HTTP Status Code. Lihat menu Status Codes di GoAccess. Apakah banyak error 404? Apa artinya bagi server?
Task 4	User-Agent (browser, bot, dll). Lihat menu User Agents di GoAccess. Apakah ada User-Agent mencurigakan (contoh: sqlmap/1.4, nikto/2.1.6)?
Task 5	Deteksi Potensi Serangan dengan menggunakan perintah grep untuk mencari pola serangan: <pre>➤ sudo zgrep -Ei "union select drop <script> alert\(wp-login phpmyadmin" /var/log/apache2/access.log*</pre> IP mana yang mencoba SQL Injection/XSS? Bagaimana cara kita memblokir IP tersebut?

Kemukakan Pendapat Anda:

Apakah server sedang diserang? Apa buktinya?

Command Tools #3

Monitoring Apache Access Using OpenAI LLM, WhatsApp Gateway & Jenkins

Ubuntu Server 22.04

Apache is part of the **LAMP stack** of software for **Linux (Linux, Apache, MySQL, PHP)**. Apache is responsible for serving web pages to people looking at our website.

Apache Access Log

Apache uses the **access log files** to record information about every visitor to our site. We can see which files visitors view, how the web server responds to requests, and other information such as the web browsers visitors use.

Location Server Debian or Ubuntu:

/var/log/apache2/access.log

Type Ubuntu Command :

- *sudo su*
- *cat /var/log/apache2/access.log*
- *nano /var/log/apache2/access.log*

Apache Error Log

- ❖ The **apache error log** is where information is recorded about any **errors** or anomalies it encounters.
- ❖ Many of the “errors” Apache records are typically *minor*, such as a **visitor requesting a file that doesn't exist**.
- ❖ Apache also uses the **error logs to record warnings** that can indicate a potential problem with a particular event or configuration.

Location Server Debian or Ubuntu:

/var/log/apache2/error.log

Type Ubuntu Command :

- *sudo su*
- *cat /var/log/apache2/error.log*
- *nano /var/log/apache2/error.log*

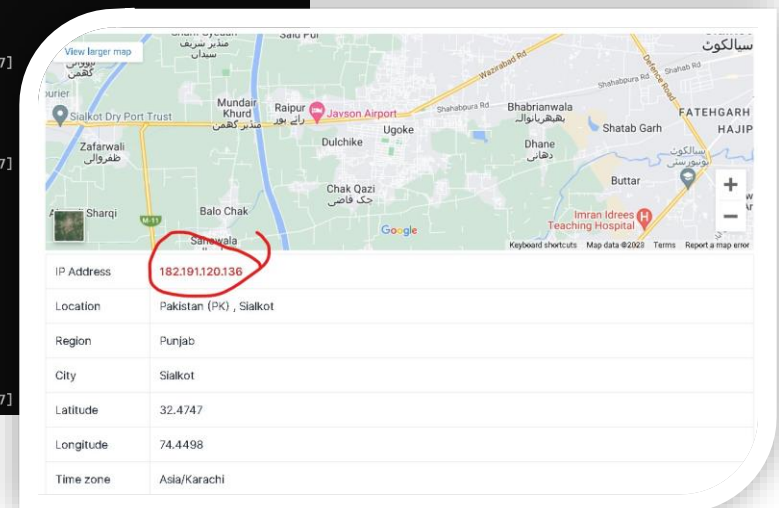
Monitor file error.log Apache secara real-time:

- *tail -f /var/log/apache2/error.log*

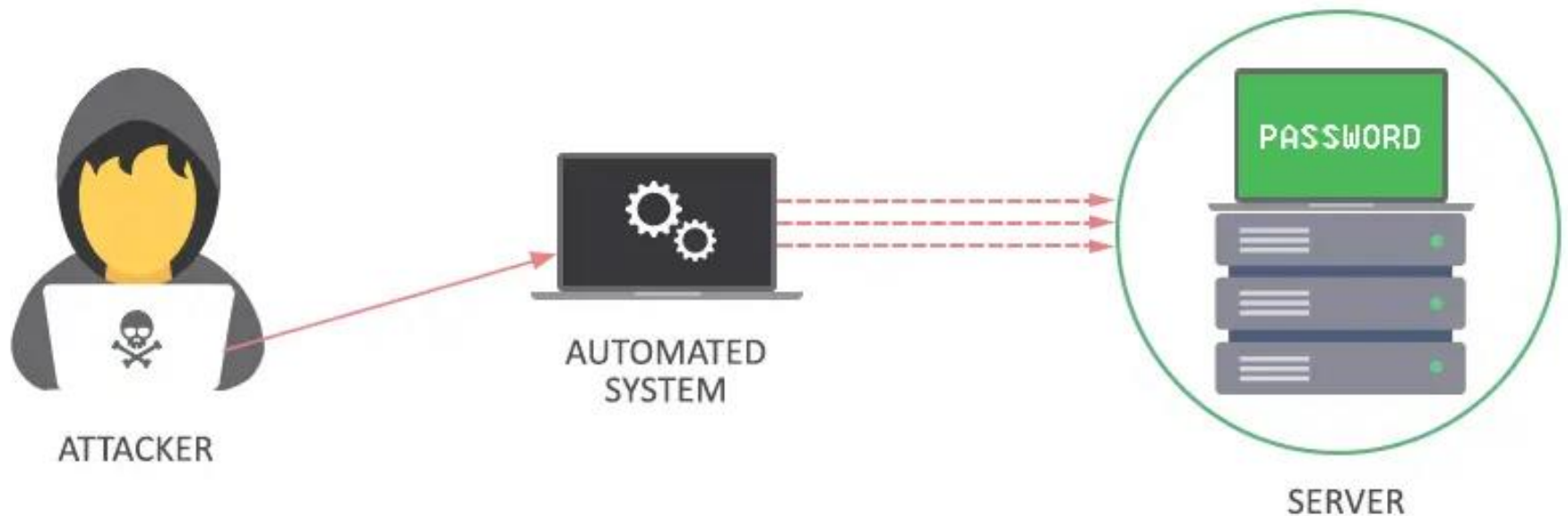
Lakukan monitoring pada Server Kalian:

- sudo su
- sudo cat /var/log/auth.log | grep "denied"
- sudo cat /var/log/auth.log | grep "192.3.228.147" # IP address tersangka
- sudo cat /var/log/auth.log | grep "Accepted publickey"
- sudo cat /var/log/auth.log | grep "Connection closed by authenticating"
- sudo cat /var/log/auth.log | grep "Connection closed by invalid user"
- sudo zgrep "Failed password" /var/log/auth.log*
- sudo zgrep "204.76.203.28" /var/log/auth.log*
- sudo zgrep -E "Failed password for (admin|root|sql)" /var/log/auth.log*
- sudo zgrep -E "Received disconnect from" /var/log/auth.log*

```
semmy@presensi: ~  
Sep 16 00:07:51 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:07:54 presensi phpMyAdmin[18162]: message repeated 4 times: [ user denied: root (mysql-denied) from 192.3.228.147]  
Sep 16 00:07:55 presensi phpMyAdmin[18162]: user denied: phpmyadmin (mysql-denied) from 192.3.228.147  
Sep 16 00:07:55 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:07:56 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:07:56 presensi phpMyAdmin[18162]: user denied: admin (mysql-denied) from 192.3.228.147  
Sep 16 00:07:57 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:07:57 presensi phpMyAdmin[18162]: user denied: web (mysql-denied) from 192.3.228.147  
Sep 16 00:07:58 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:07:59 presensi phpMyAdmin[18162]: message repeated 3 times: [ user denied: root (mysql-denied) from 192.3.228.147]  
Sep 16 00:08:00 presensi phpMyAdmin[18162]: user denied: wordpress (mysql-denied) from 192.3.228.147  
Sep 16 00:08:00 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:02 presensi phpMyAdmin[18162]: message repeated 3 times: [ user denied: root (mysql-denied) from 192.3.228.147]  
Sep 16 00:08:02 presensi phpMyAdmin[18162]: user denied: developer (mysql-denied) from 192.3.228.147  
Sep 16 00:08:03 presensi phpMyAdmin[18162]: user denied: dbs (mysql-denied) from 192.3.228.147  
Sep 16 00:08:03 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:04 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:04 presensi phpMyAdmin[18162]: user denied: wp (mysql-denied) from 192.3.228.147  
Sep 16 00:08:05 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:09 presensi phpMyAdmin[18162]: message repeated 8 times: [ user denied: root (mysql-denied) from 192.3.228.147]  
Sep 16 00:08:09 presensi phpMyAdmin[18162]: user denied: newuser (mysql-denied) from 192.3.228.147  
Sep 16 00:08:10 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:10 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:11 presensi phpMyAdmin[18162]: user denied: database (mysql-denied) from 192.3.228.147  
Sep 16 00:08:11 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:12 presensi phpMyAdmin[18162]: message repeated 2 times: [ user denied: root (mysql-denied) from 192.3.228.147]  
Sep 16 00:08:13 presensi phpMyAdmin[18162]: user denied: user (mysql-denied) from 192.3.228.147  
Sep 16 00:08:14 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:15 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:16 presensi phpMyAdmin[18162]: user denied: newuser (mysql-denied) from 192.3.228.147  
Sep 16 00:08:16 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:17 presensi phpMyAdmin[18162]: user denied: dev (mysql-denied) from 192.3.228.147  
Sep 16 00:08:17 presensi phpMyAdmin[18162]: user denied: admin (mysql-denied) from 192.3.228.147  
Sep 16 00:08:18 presensi phpMyAdmin[18162]: user denied: admin (mysql-denied) from 192.3.228.147  
Sep 16 00:08:19 presensi phpMyAdmin[18162]: user denied: admin (mysql-denied) from 192.3.228.147  
Sep 16 00:08:20 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:20 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:21 presensi phpMyAdmin[18162]: user denied: phpmyadmin (mysql-denied) from 192.3.228.147  
Sep 16 00:08:22 presensi phpMyAdmin[18162]: user denied: root (mysql-denied) from 192.3.228.147  
Sep 16 00:08:26 presensi phpMyAdmin[18162]: message repeated 7 times: [ user denied: root (mysql-denied) from 192.3.228.147]
```

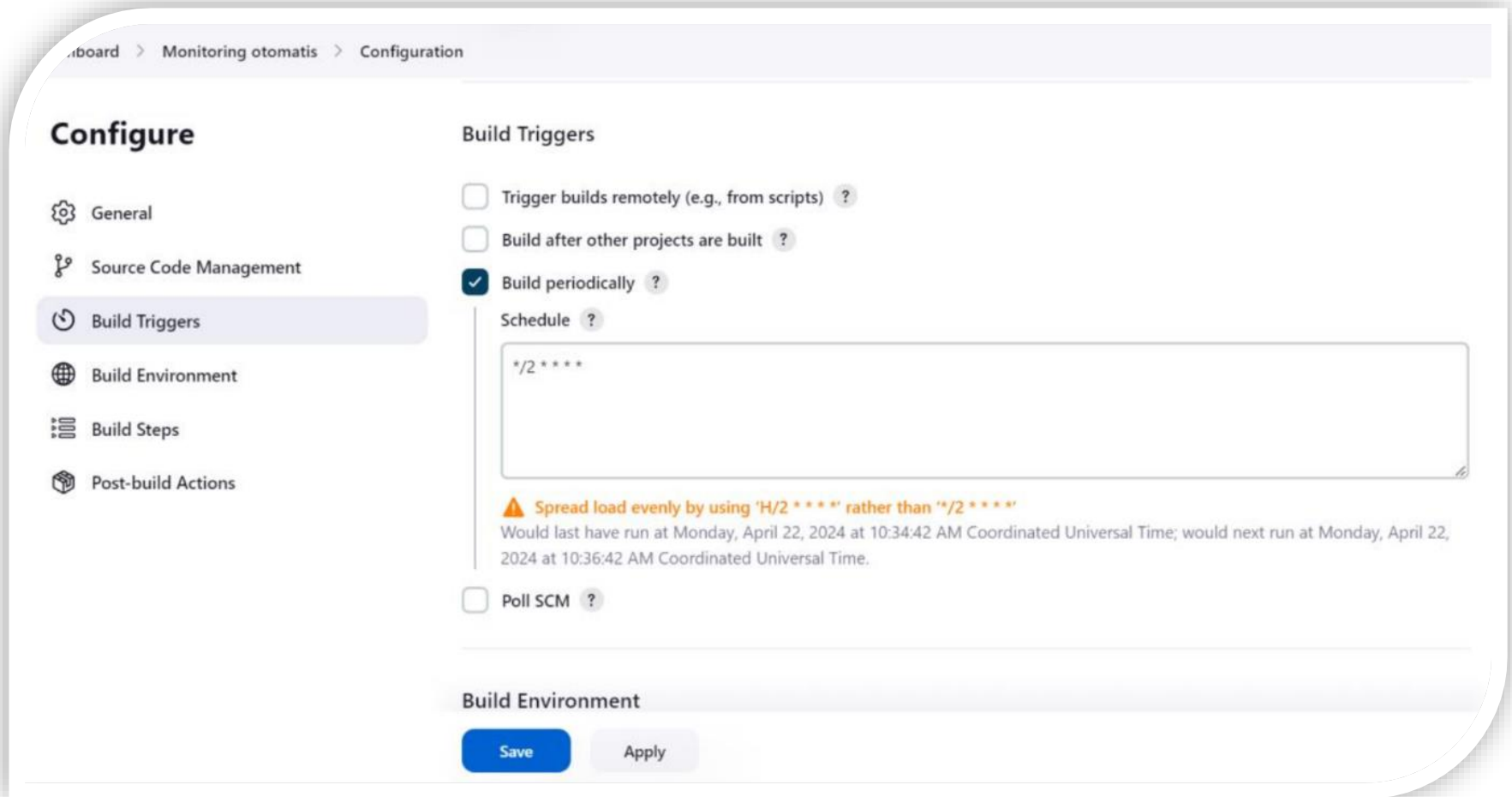


Dari file log berikut ini, identifikasi tipe serang berikut. Menurut anda bagaimana cara untuk mencegah tipe serangan seperti diatas?



Brute Force Attack adalah metode untuk melakukan membobol sebuah akun dengan cara menebak *username* dan bahkan *password* yang digunakan oleh users (*pengguna*) tersebut. Metode ini akan dilakukan dengan berbagai percobaan melalui *trial* and *error* hingga akhirnya berhasil untuk masuk ke dalam akun yang dituju.

Jelaskan 2 tipe serangan keamanan
XSS (Cross-Site Scripting) dan
CSRF (Cross-Site Request Forgery) berikut?
Bagaimana cara anda untuk menangani tipe
serangan seperti ini?



Pada bagian Build triggers, centang Build periodically kemudian isi cron syntax pada schedule untuk mengatur frekuensi job dijalankan, dalam kasus ini saya akan membuat job dijalankan setiap 2 menit dengan perintah `*/2 * * * *`.

Configure

Build Steps

- General
- Source Code Management
- Build Triggers
- Build Environment
- Build Steps**
- Post-build Actions

Execute shell ?

Command

See [the list of available environment variables](#)

```
cat /var/log/auth.log | grep "invalid"

echo =====

cat /var/log/apache2/access.log
```

Advanced ▾

Add build step ▾

Pada bagian Build steps,

- 1) Click Add build steps lalu pilih Execute shell. Di dalamnya kita akan memasukan perintah untuk menampilkan isi file auth.log pada path `/var/log/auth.log` dan file access.log pada path `/var/log/apache2/access.log`. Pada code di bawah saya menambahkan **grep** **"invalid"** agar log yang ditampilkan dari `auth.log` hanya yang mengandung kata invalid. Setelah sudah memasukan perintah kita klik apply dan save.
- 2) Perintah **grep (Global Regular Expression Print)** berfungsi untuk mencari baris-baris dalam sebuah file atau output yang mengandung pola (*pattern*) tertentu. Jadi, perintah berikut ini artinya (**| grep "invalid"**) menyaring hasil untuk hanya menampilkan baris-baris yang mengandung kata **"invalid"**.

board > Monitoring otomatis > #2 > Console Output

Status

</> Changes

Console Output

View as plain text

Edit Build Information

Delete build '#2'

← Previous Build

Console Output

```
Started by timer
Running as SYSTEM
Building in workspace /root/.jenkins/workspace/Monitoring otomatis
[Monitoring otomatis] $ /bin/sh -xe /tmp/jenkins8758426560395266642.sh
+ cat /var/log/auth.log
+ grep invalid
Apr 16 02:49:14 ovel-server sshd[2174]: Failed password for invalid user zabbix from 68.183.225.172 port 52636
ssh2
Apr 16 02:49:16 ovel-server sshd[2174]: Connection closed by invalid user zabbix 68.183.225.172 port 52636
[preauth]
Apr 16 02:49:23 ovel-server sshd[2178]: Failed password for invalid user support from 68.183.225.172 port 54564
ssh2
Apr 16 02:49:24 ovel-server sshd[2178]: Connection closed by invalid user support 68.183.225.172 port 54564
[preauth]
Apr 16 02:49:26 ovel-server sshd[2180]: Failed password for invalid user alan from 68.183.225.172 port 55290 ssh2
Apr 16 02:49:27 ovel-server sshd[2180]: Connection closed by invalid user alan 68.183.225.172 port 55290 [preauth]
Apr 16 02:49:32 ovel-server sshd[2182]: Failed password for invalid user default from 68.183.225.172 port 55740
ssh2
Apr 16 02:49:33 ovel-server sshd[2182]: Connection closed by invalid user default 68.183.225.172 port 55740
[preauth]
Apr 16 02:49:50 ovel-server sshd[2192]: Failed password for invalid user test1 from 68.183.225.172 port 59744 ssh2
Apr 16 02:49:51 ovel-server sshd[2192]: Connection closed by invalid user test1 68.183.225.172 port 59744
```

Output yang akan ditampilkan !!!

Opsi #1:

Next Report !!!

1. Buatlah demo Automated monitoring (focus pada keamanan) menggunakan Jenkins. tidak ada Batasan, kalian bisa gunakan Teknik yang baru atau penggunaan tools yang berbeda.
2. Next meeting students akan diminta untuk melakukan Demo Automated monitoring keamanan system di dalam class !!!

Opsi #2:

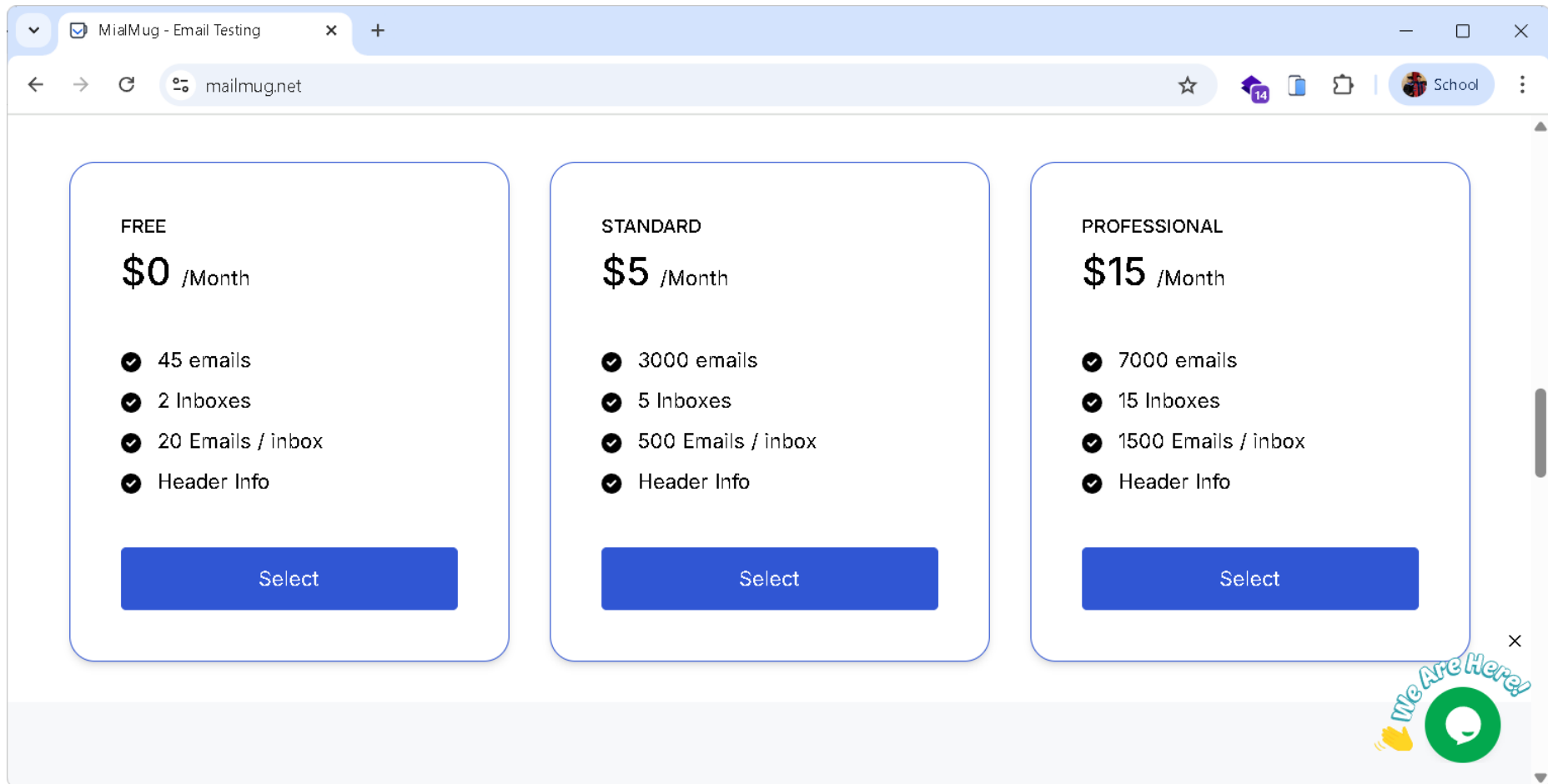
Challenges *for* students !!!

Menggabungkan LLM seperti GPT-4o atau Gemini 2.0 Flash, WhatsApp Gateway (Fonnte), Mail Sandbox (MailMug), Mail Gateway (MailGun), dan Jenkins untuk membuat **sistem monitoring otomatis** di server Ubuntu 22.04. Sistem sederhana ini dikhususkan untuk mendeteksi percobaan login paksa (*brute-force*) yang mungkin dilakukan dari sisi client SSH dan bahkan bisa menjelaskan/memberi rekomendasi yang *smart* untuk admin.

- ✓ OpenAI GPT-4o atau Gemini 2.0 Flash → buat *analisis cerdas*
- ✓ WhatsApp Gateway via Fonnte → buat *notifikasi real-time*
- ✓ Jenkins → untuk *otomasi periodik*

MailMug (SMTP Sandbox)

Tujuan utama untuk pengujian (*testing*) pengiriman email dari aplikasi, tanpa benar-benar mengirim ke end-users, digunakan di development / staging server.. Contoh service yaitu MailMug, MailHog, Mailtrap dan Papercut SMTP.

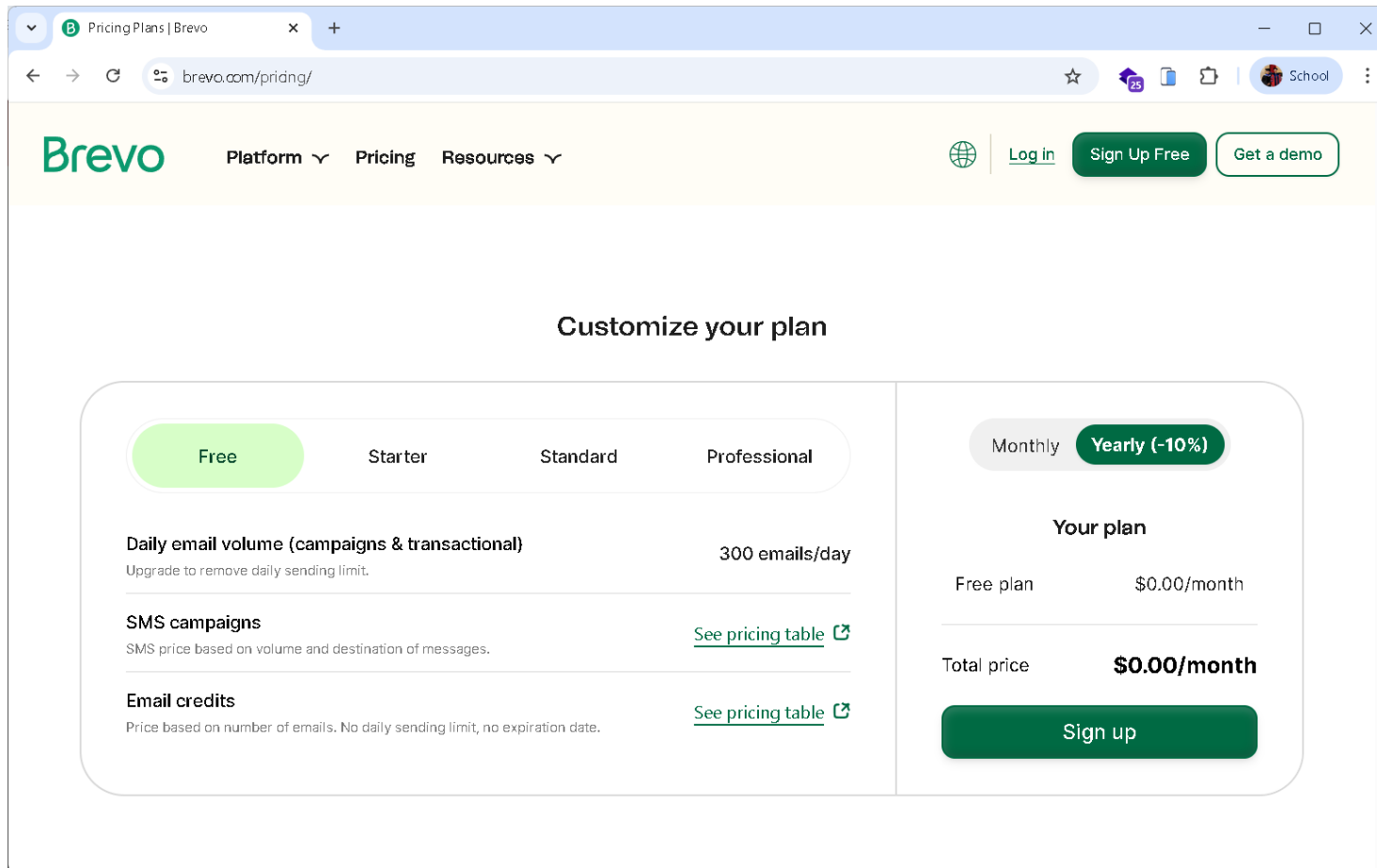


The screenshot shows the MailMug website in a web browser. The browser's address bar displays 'mailmug.net'. The website features three pricing plans, each with a 'Select' button. A 'We Are Here!' chat bubble is visible in the bottom right corner.

Plan	Price /Month	Features
FREE	\$0	✓ 45 emails ✓ 2 Inboxes ✓ 20 Emails / inbox ✓ Header Info
STANDARD	\$5	✓ 3000 emails ✓ 5 Inboxes ✓ 500 Emails / inbox ✓ Header Info
PROFESSIONAL	\$15	✓ 7000 emails ✓ 15 Inboxes ✓ 1500 Emails / inbox ✓ Header Info

Mailgun (Mail Gateway)

Tujuan utama untuk mengirim, memfilter, dan mengamankan email keluar/masuk dari sistem ke internet, digunakan di production (server nyata). Contoh service yaitu Postfix, Exim, Sendmail, Microsoft Exchange, Mailgun dan AWS SES Gateway.



The screenshot shows the Brevo Pricing Plans page in a web browser. The browser's address bar shows 'brevo.com/pricing/'. The page has a yellow header with the Brevo logo, navigation links for Platform, Pricing, and Resources, and buttons for Log in, Sign Up Free, and Get a demo. The main content area is titled 'Customize your plan' and features a plan selector with options: Free (selected), Starter, Standard, and Professional. Below the selector, details for the Free plan are shown: 'Daily email volume (campaigns & transactional)' with a limit of 300 emails/day, 'SMS campaigns' with a link to the pricing table, and 'Email credits' with a link to the pricing table. On the right, a summary section titled 'Your plan' shows the 'Monthly' and 'Yearly (-10%)' options, with the 'Free plan' listed at '\$0.00/month'. The 'Total price' is also '\$0.00/month', and a 'Sign up' button is at the bottom.

Brevo Pricing Plans | Brevo

brevo.com/pricing/

Brevo Platform Pricing Resources

Log in Sign Up Free Get a demo

Customize your plan

Free Starter Standard Professional

Daily email volume (campaigns & transactional)
Upgrade to remove daily sending limit. 300 emails/day

SMS campaigns
SMS price based on volume and destination of messages. [See pricing table](#)

Email credits
Price based on number of emails. No daily sending limit, no expiration date. [See pricing table](#)

Your plan

Monthly **Yearly (-10%)**

Free plan \$0.00/month

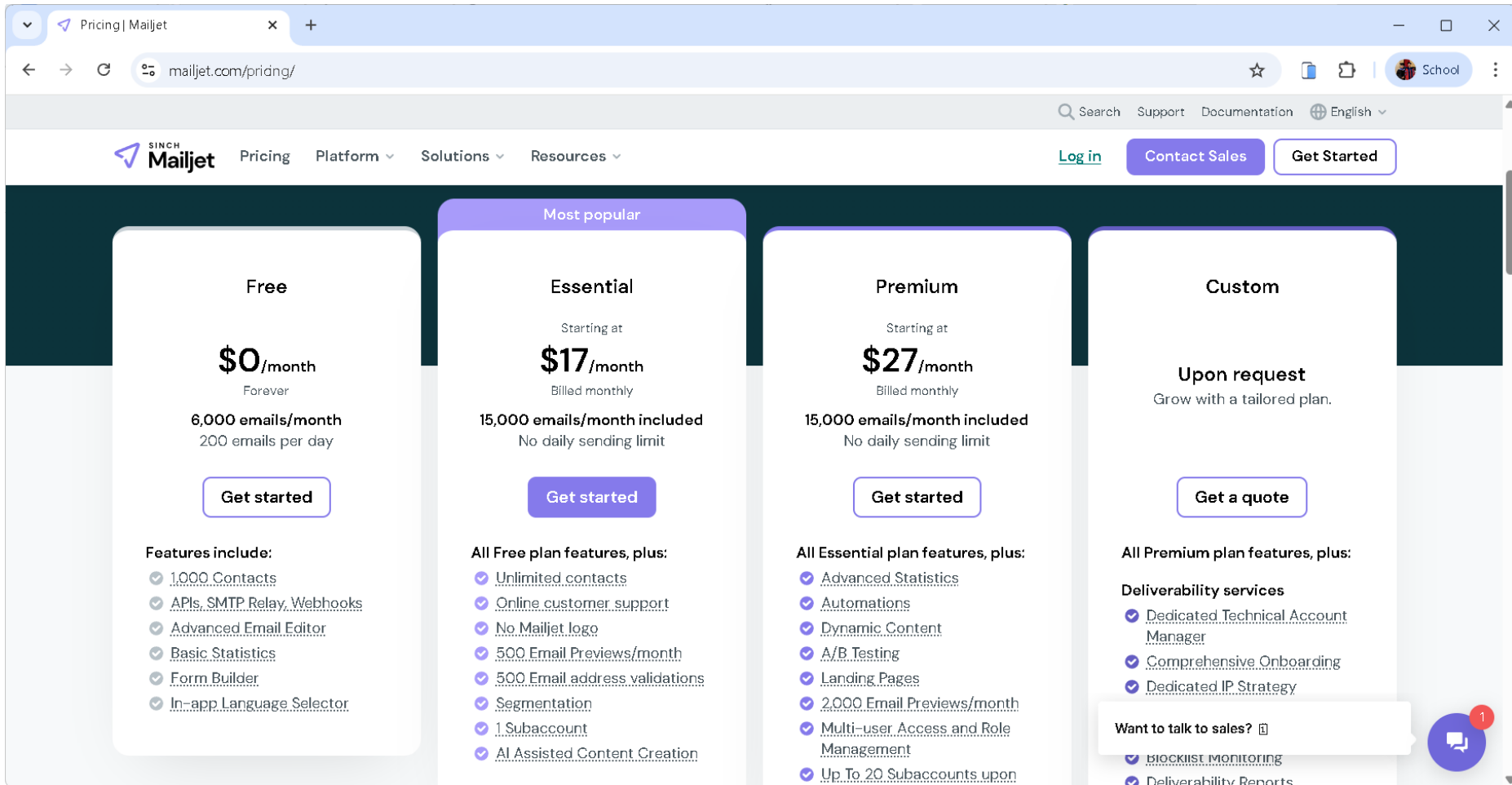
Total price **\$0.00/month**

Sign up

Mailjet (Mail Gateway)

Official website:

<https://www.mailjet.com/pricing/>



The screenshot shows the Mailjet Pricing page in a web browser. The browser's address bar displays 'mailjet.com/pricing/'. The page features a navigation bar with links to 'Pricing', 'Platform', 'Solutions', and 'Resources', along with 'Log in', 'Contact Sales', and 'Get Started' buttons. The main content area is divided into four pricing plans: Free, Essential, Premium, and Custom. Each plan is presented in a white card with a dark header. The 'Essential' plan is highlighted as 'Most popular'. The 'Free' plan offers 6,000 emails/month forever. The 'Essential' plan starts at \$17/month and includes 15,000 emails/month. The 'Premium' plan starts at \$27/month and includes 15,000 emails/month. The 'Custom' plan is available upon request. Each plan has a 'Get started' or 'Get a quote' button. Below the plan cards, there are sections for 'Features include:' for each plan, listing various capabilities like contacts, APIs, email editor, statistics, form builder, and segmentation. A chat bubble in the bottom right corner asks 'Want to talk to sales?' and shows a notification count of 1.

Pricing | Mailjet

mailjet.com/pricing/

Search Support Documentation English

Log in Contact Sales Get Started

Free

Starting at

\$0/month

Forever

6,000 emails/month
200 emails per day

Get started

Features include:

- ✓ 1,000 Contacts
- ✓ APIs, SMTP Relay, Webhooks
- ✓ Advanced Email Editor
- ✓ Basic Statistics
- ✓ Form Builder
- ✓ In-app Language Selector

Essential

Starting at

\$17/month

Billed monthly

15,000 emails/month included
No daily sending limit

Get started

All Free plan features, plus:

- ✓ Unlimited contacts
- ✓ Online customer support
- ✓ No Mailjet logo
- ✓ 500 Email Previews/month
- ✓ 500 Email address validations
- ✓ Segmentation
- ✓ 1 Subaccount
- ✓ AI Assisted Content Creation

Premium

Starting at

\$27/month

Billed monthly

15,000 emails/month included
No daily sending limit

Get started

All Essential plan features, plus:

- ✓ Advanced Statistics
- ✓ Automations
- ✓ Dynamic Content
- ✓ A/B Testing
- ✓ Landing Pages
- ✓ 2,000 Email Previews/month
- ✓ Multi-user Access and Role Management
- ✓ Up To 20 Subaccounts upon request

Custom

Starting at

Upon request

Grow with a tailored plan.

Get a quote

All Premium plan features, plus:

Deliverability services

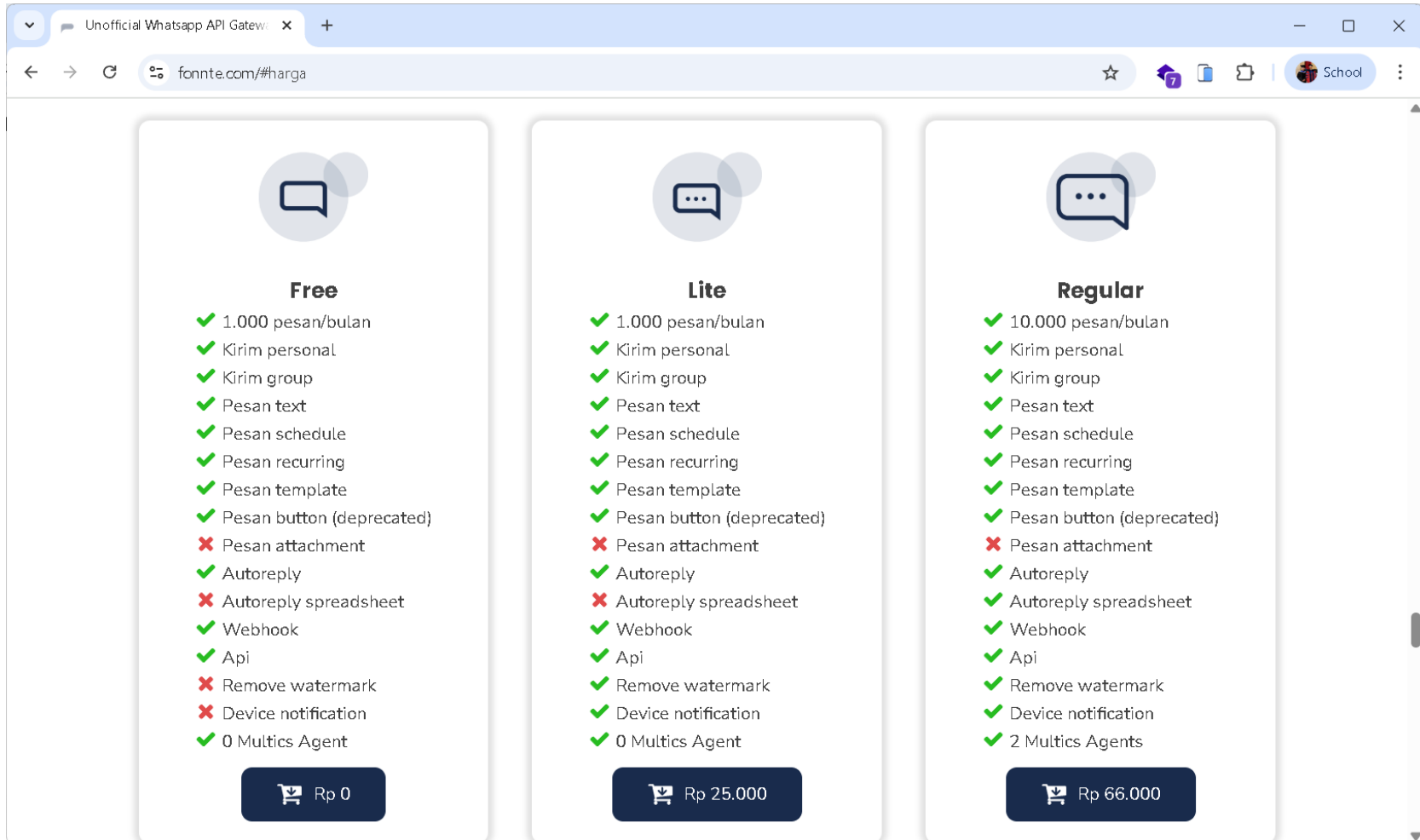
- ✓ Dedicated Technical Account Manager
- ✓ Comprehensive Onboarding
- ✓ Dedicated IP Strategy

Want to talk to sales?

- ✓ Blacklist Monitoring
- ✓ Deliverability Reports

Fonnte (WhatsApp Gateway)

Fonnte adalah WhatsApp Gateway API berbasis cloud, buatan Indonesia, yang memungkinkan mengirim dan menerima pesan WhatsApp secara otomatis dari sistem atau server sendiri.



The screenshot shows the Fonnte website in a web browser. The browser's address bar displays 'fonnte.com/#harga'. The website features three pricing plans, each with a list of features and a price tag at the bottom.

Plan	Features	Price
Free	✓ 1.000 pesan/bulan ✓ Kirim personal ✓ Kirim group ✓ Pesan text ✓ Pesan schedule ✓ Pesan recurring ✓ Pesan template ✓ Pesan button (deprecated) ✗ Pesan attachment ✓ Autoreply ✗ Autoreply spreadsheet ✓ Webhook ✓ Api ✗ Remove watermark ✗ Device notification ✓ 0 Multics Agent	Rp 0
Lite	✓ 1.000 pesan/bulan ✓ Kirim personal ✓ Kirim group ✓ Pesan text ✓ Pesan schedule ✓ Pesan recurring ✓ Pesan template ✓ Pesan button (deprecated) ✗ Pesan attachment ✓ Autoreply ✗ Autoreply spreadsheet ✓ Webhook ✓ Api ✓ Remove watermark ✓ Device notification ✓ 0 Multics Agent	Rp 25.000
Regular	✓ 10.000 pesan/bulan ✓ Kirim personal ✓ Kirim group ✓ Pesan text ✓ Pesan schedule ✓ Pesan recurring ✓ Pesan template ✓ Pesan button (deprecated) ✗ Pesan attachment ✓ Autoreply ✓ Autoreply spreadsheet ✓ Webhook ✓ Api ✓ Remove watermark ✓ Device notification ✓ 2 Multics Agents	Rp 66.000

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

